



DexterChem

REVISION SERIES NAMED REACTIONS



NOTE

A dedicated Revision playlist link is in the description box of this chapter.

Containing :

- Revision Video***
- 2022 PYQs***
- 2023 PYQs***

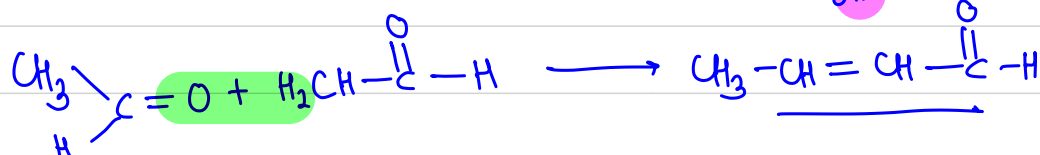
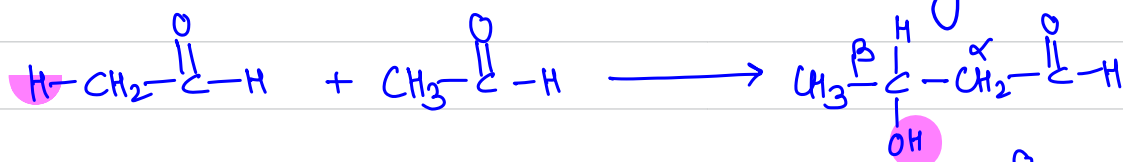
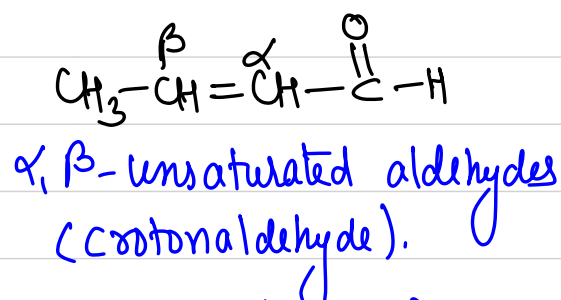
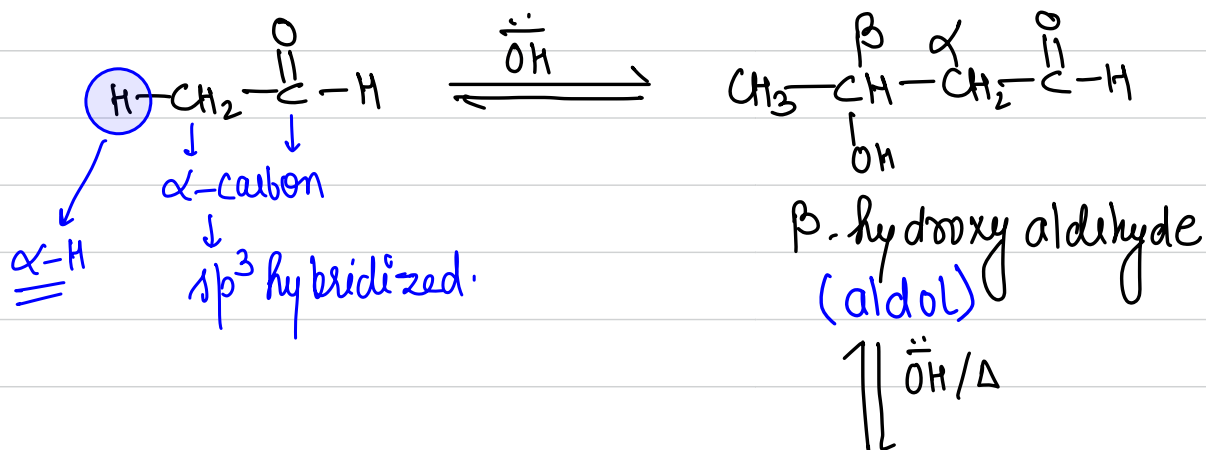
To get Notes + PYQs of this chapter

Link in description or go to www.dexterchem.in/courses to find PYQ and Revision Notes.

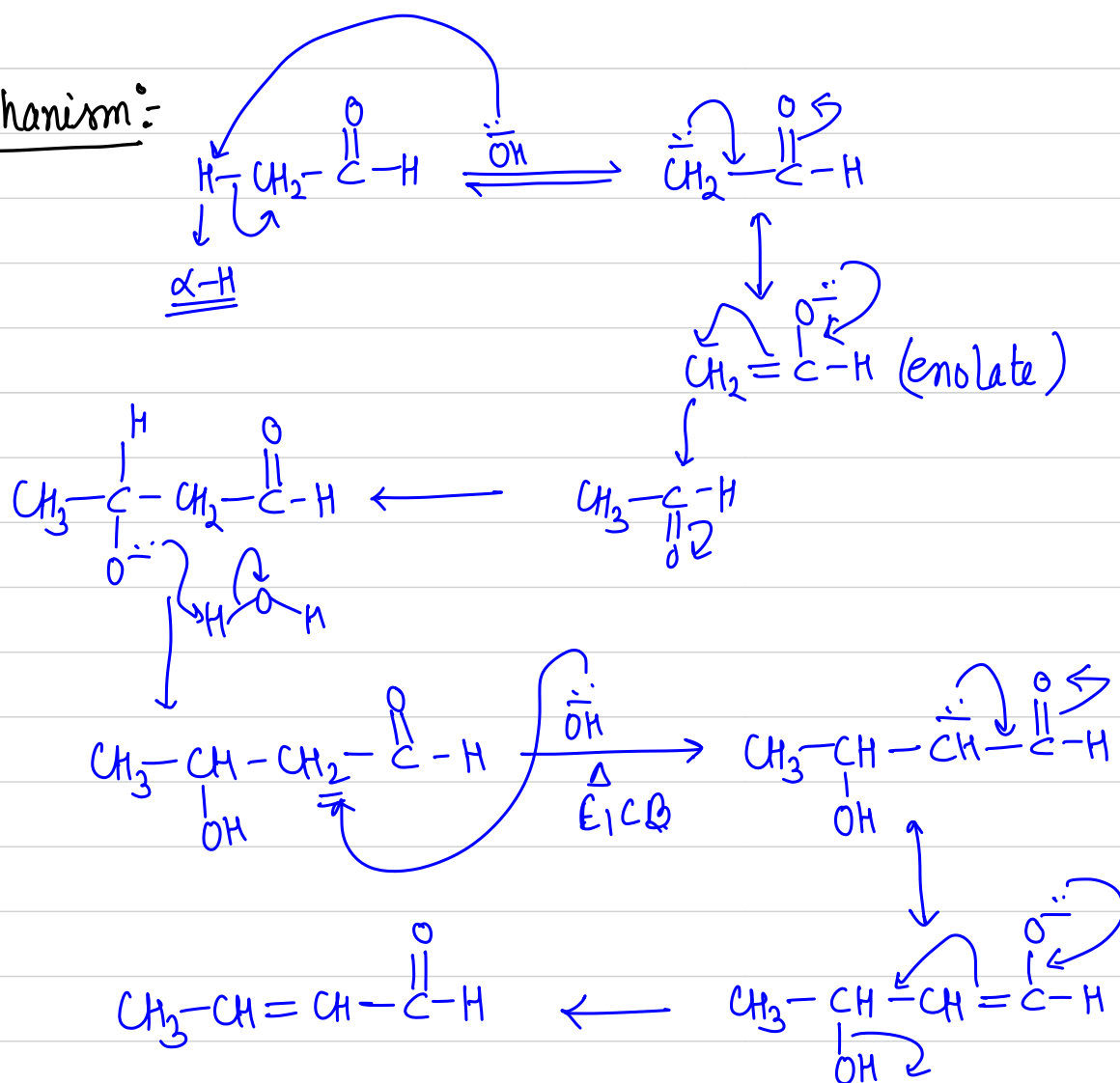
Named Reactions

① Aldol reaction and Condensation:-

Rxn of an aldehyde or ketone containing at least one α -H in presence of dil $\ddot{\text{O}}\text{H}$ to form β -hydroxy aldehydes/ketones is known as aldol reaction and further heating in presence of $\ddot{\text{O}}\text{H}$ to form α, β unsaturated aldehydes & ketones is known as aldol condensation.



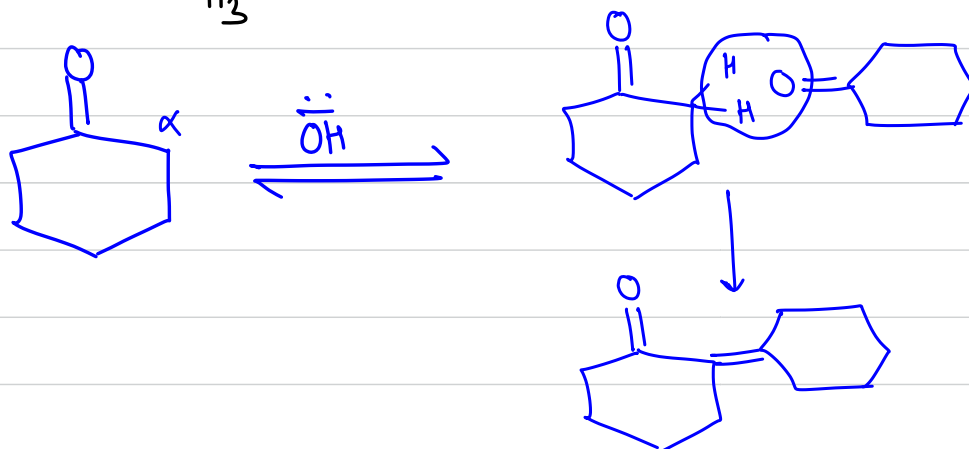
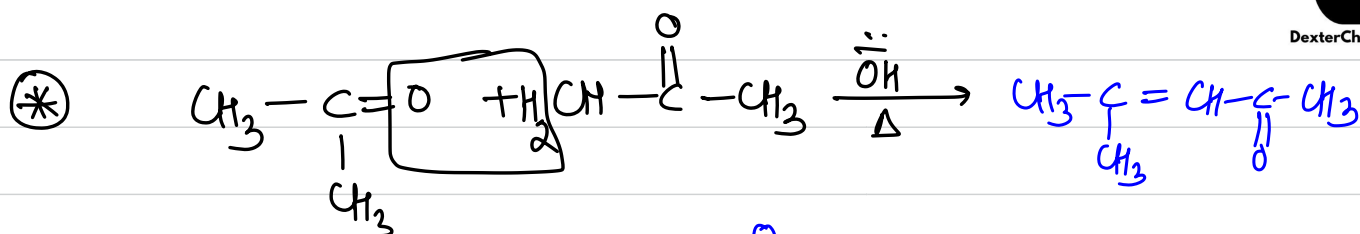
Mechanism:-



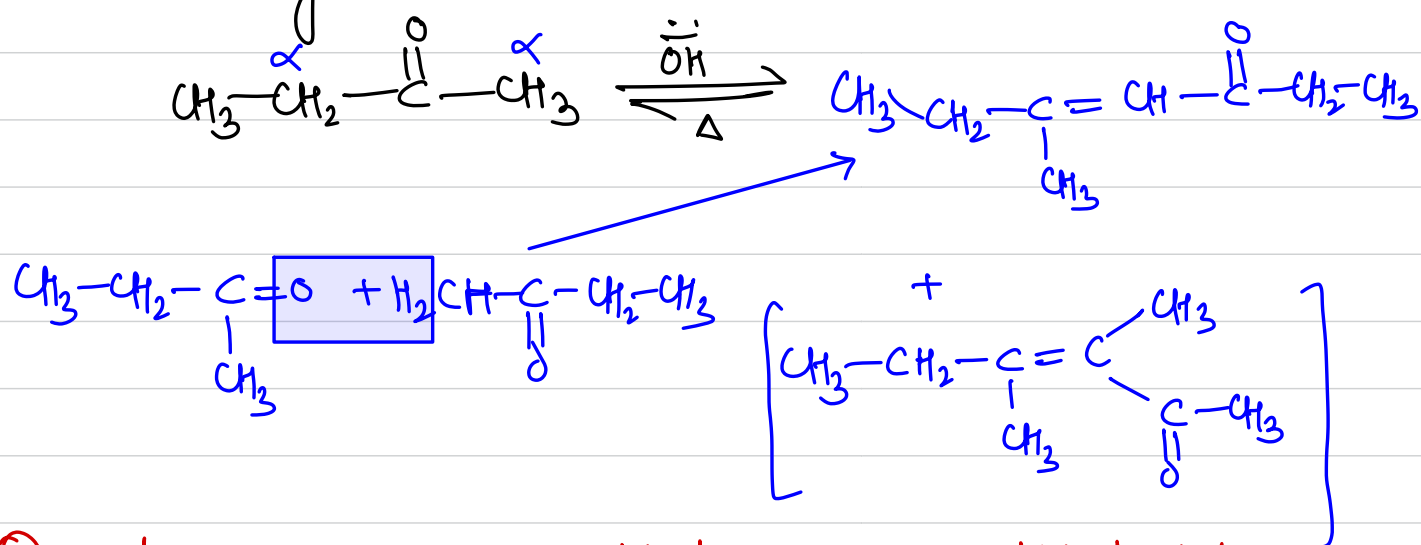
⊗ If the conc. of OH^- is less then the formation of enolate ion becomes the r.d.s. (i.e. first step).

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⊗ If the conc. of OH^- is high then the attack of enolate on carbonyl carbon becomes r.d.s.



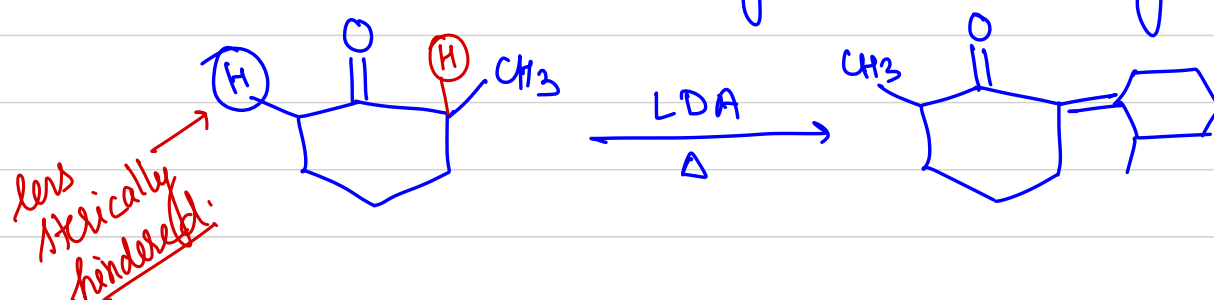
* For Unsymmetrical Ketones α -



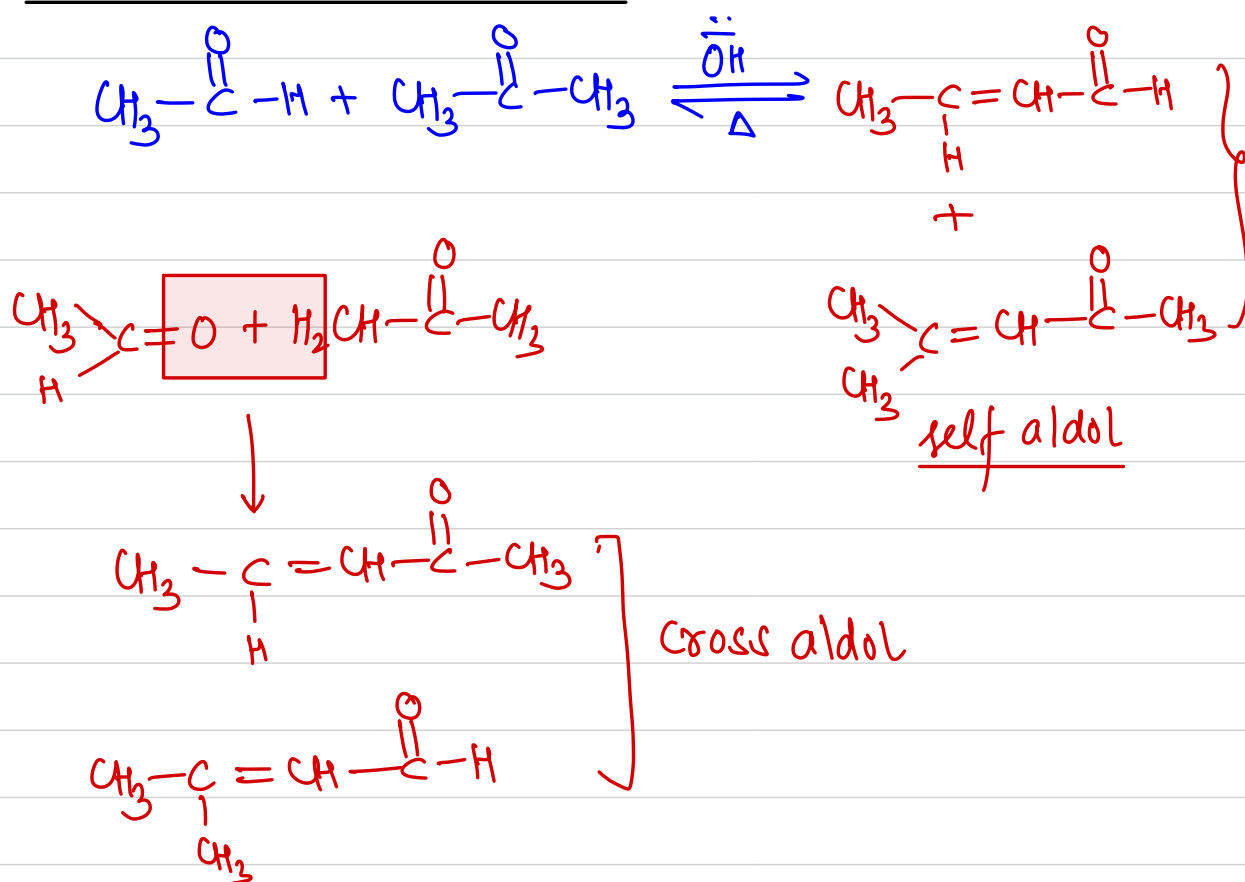
(*) In case of unsymmetrical ketones multiple products are formed. To avoid this situation, we use LDA (Bulky base), which causes directed aldol.

LDA $\rightarrow$ Lithium diisopropyl amide. $\text{Li}^+ \ddot{\text{N}}(\text{CH}_2\text{CH(CH}_3)_2)_2$

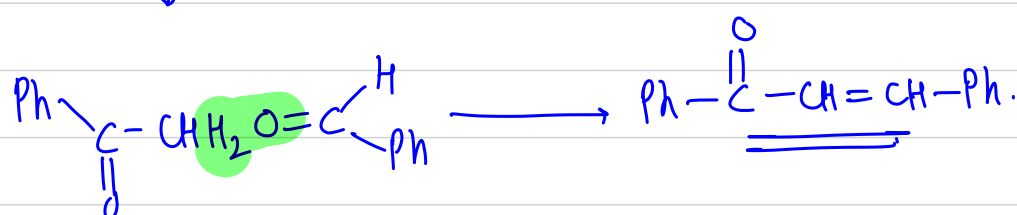
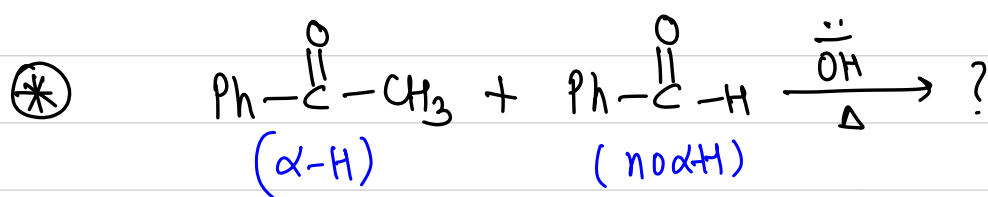
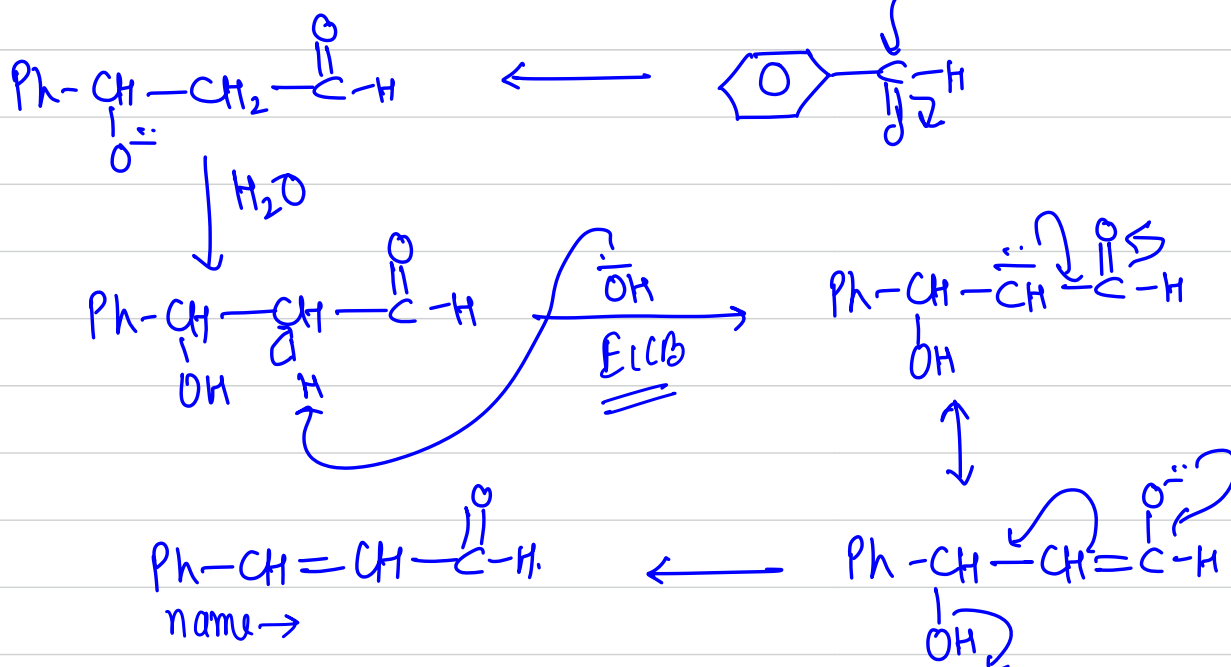
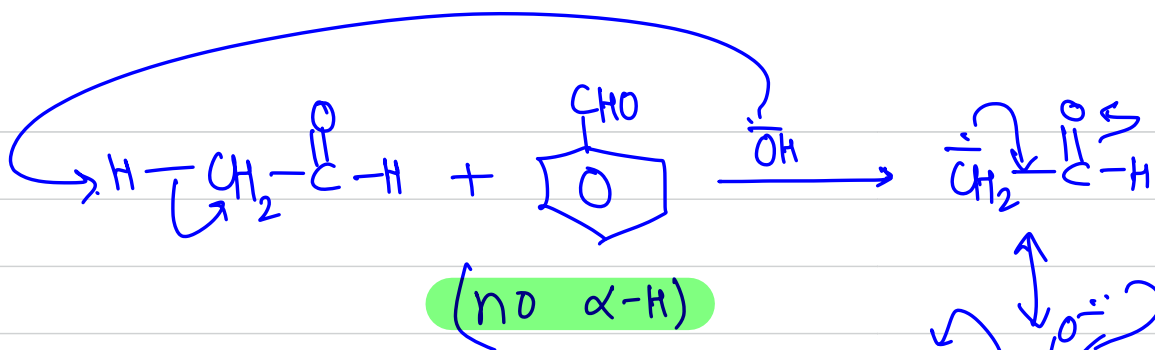
LDA will take the $[H^+]$ from less sterically hindered site.



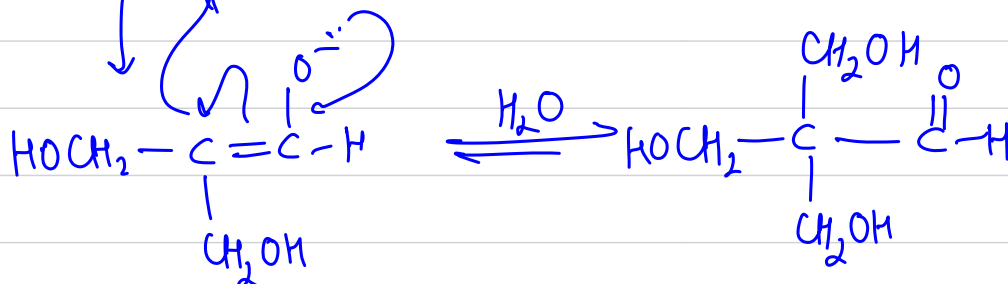
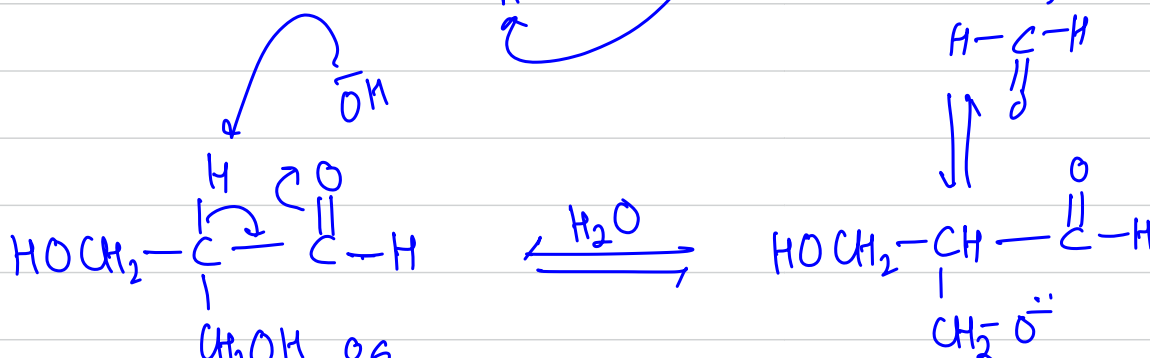
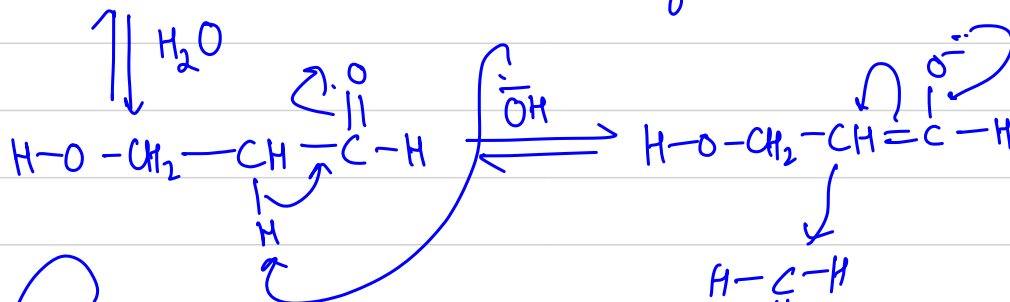
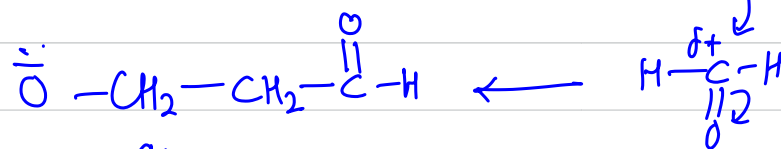
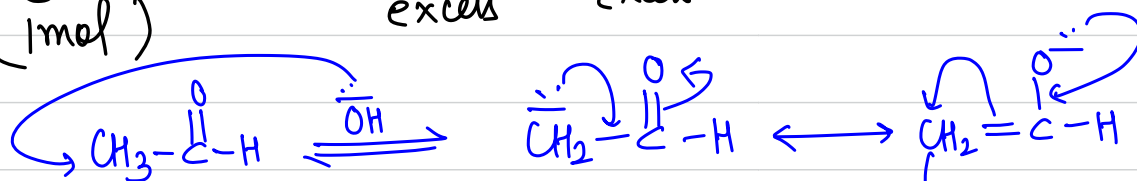
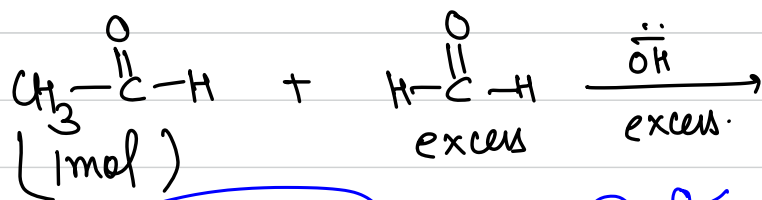
(*) Cross Aldol condensation :-



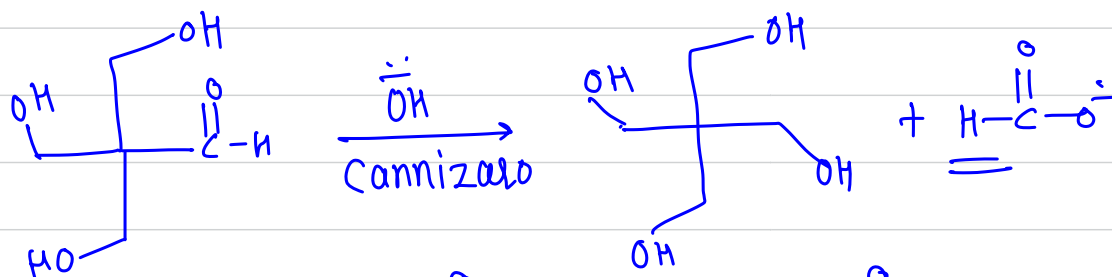
(*) For successful cross aldol condensation one partner should be enolizable (having $\alpha-H$) & the other partner should be non-enolizable (having no $\alpha-H$).



Advanced
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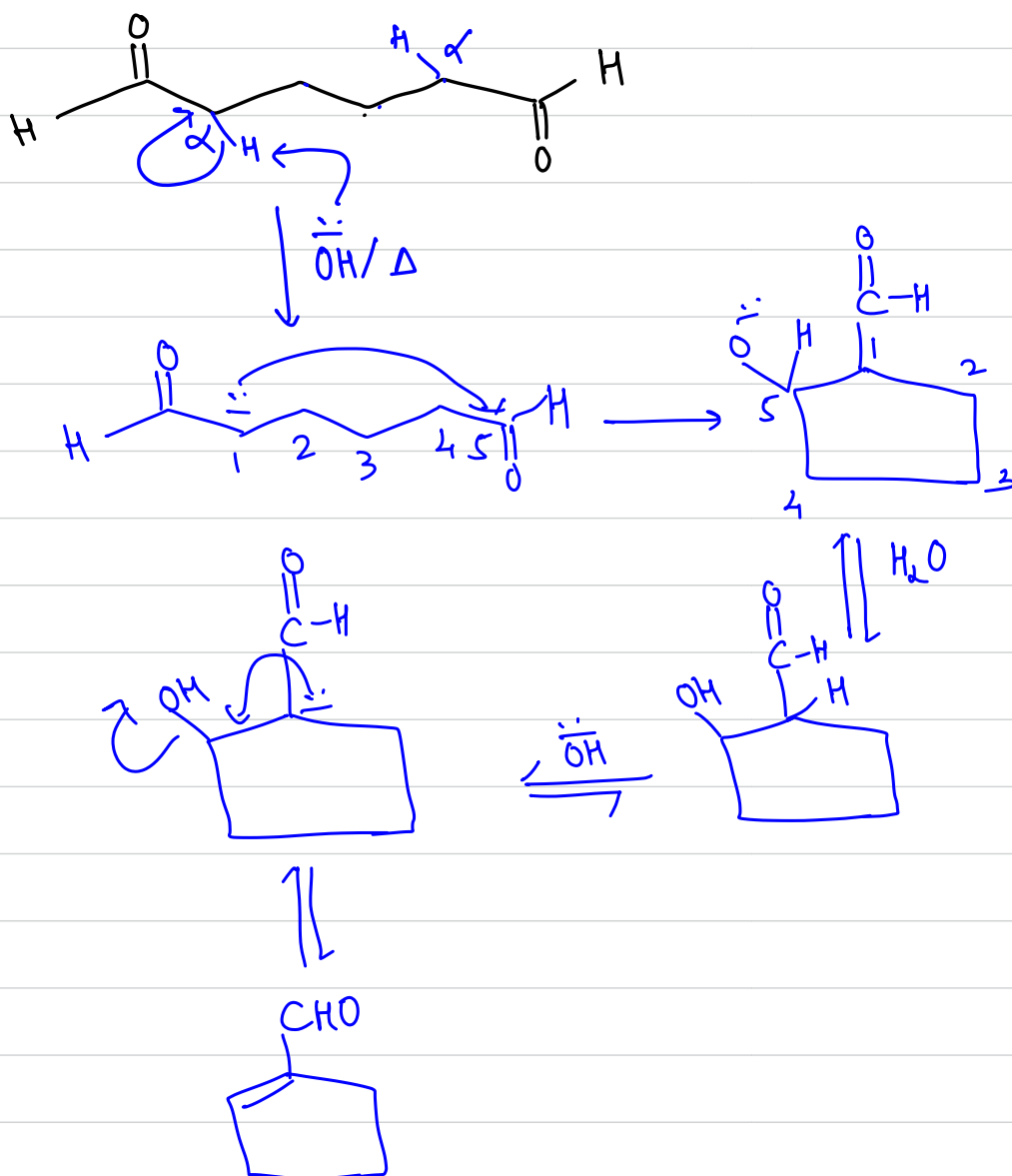


Till now 3 times aldol reaction has taken place.



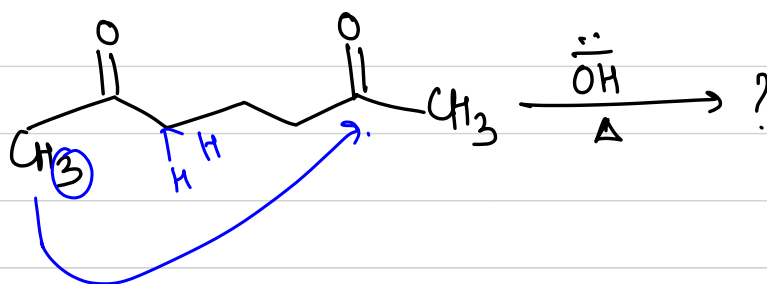
for 1 mol of $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ 4 moles of $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ are used.

(*) Intramolecular Aldol condensation:-

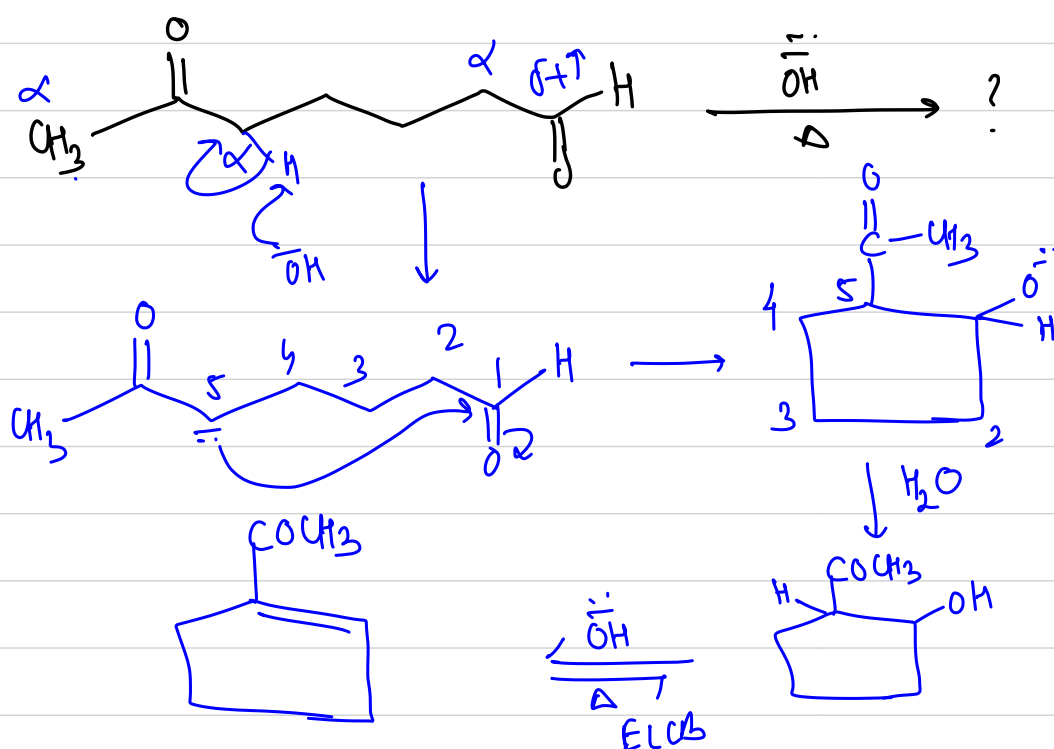


Probable Question:-

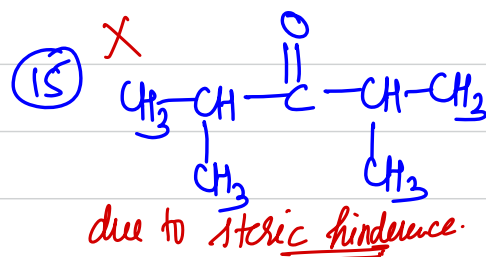
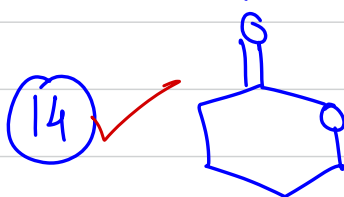
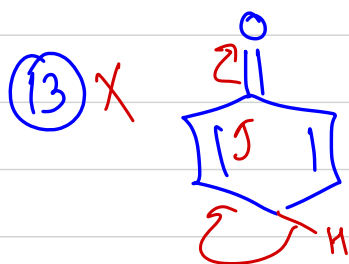
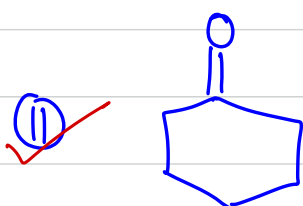
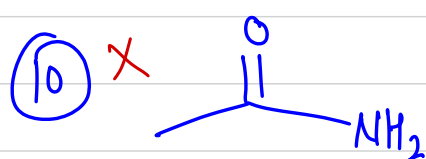
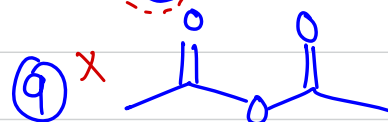
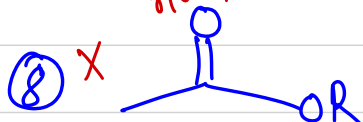
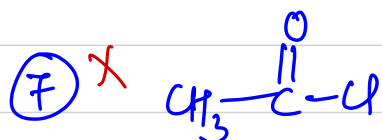
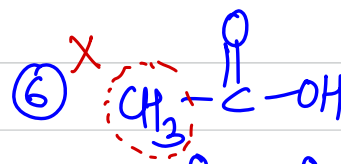
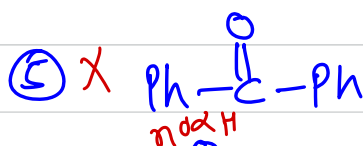
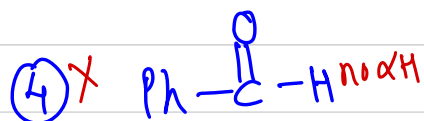
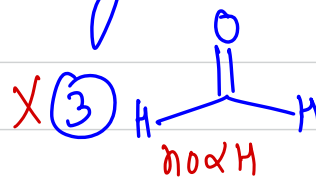
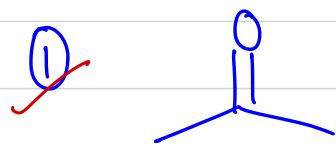
JEE MAINS:-



JEE ADVANCED:-



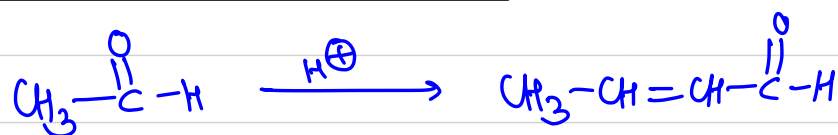
Q: which of the following will give Aldol condensation reaction:-
And also write the major product formed.



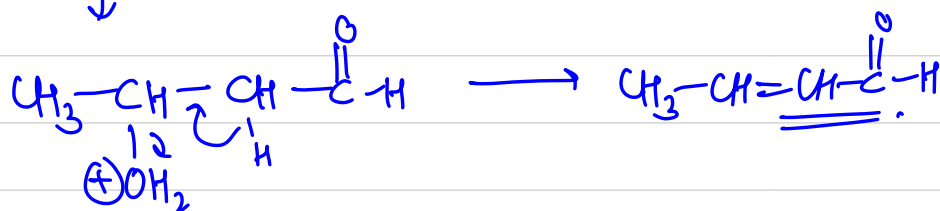
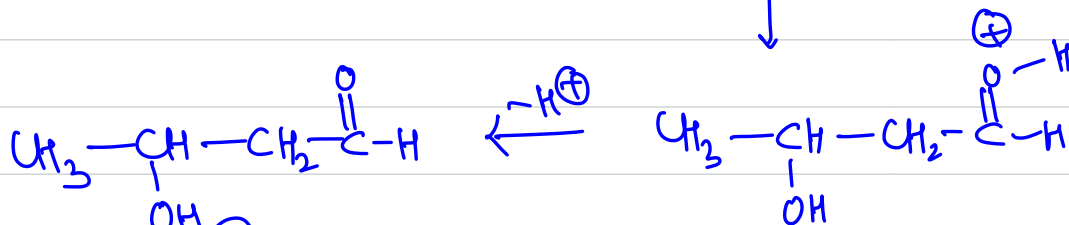
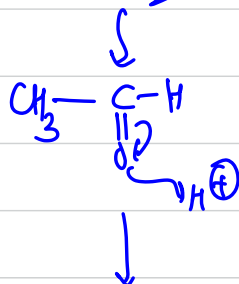
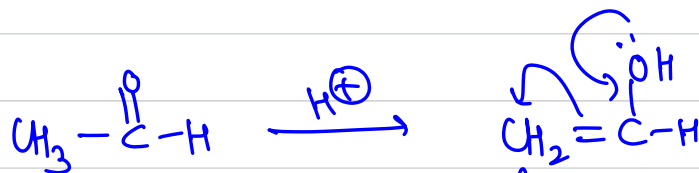
* Carboxylic acids & their derivatives do not give aldol rxn.

* cyclic esters are called lactones they give aldol rxn but at a slow rate.

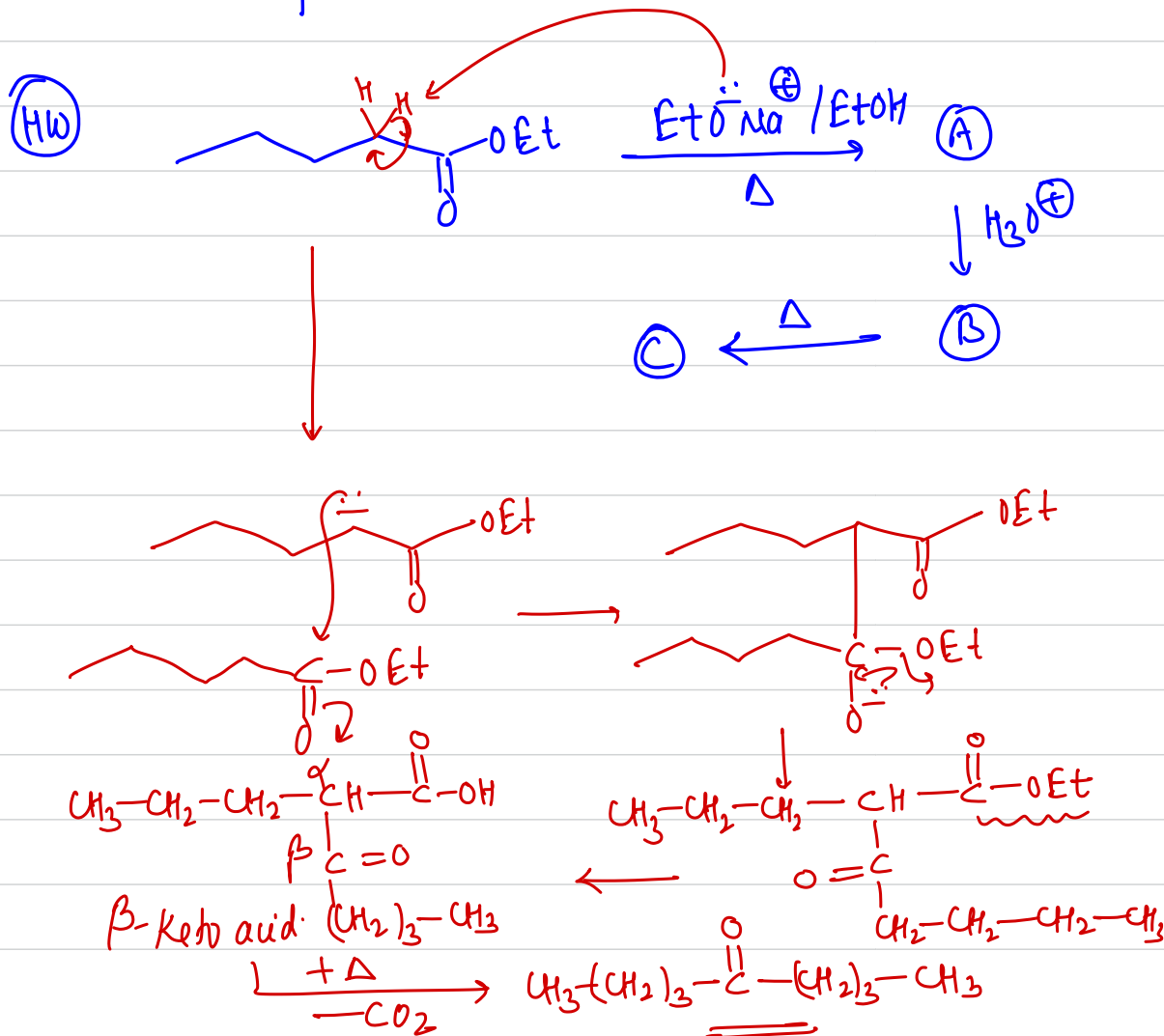
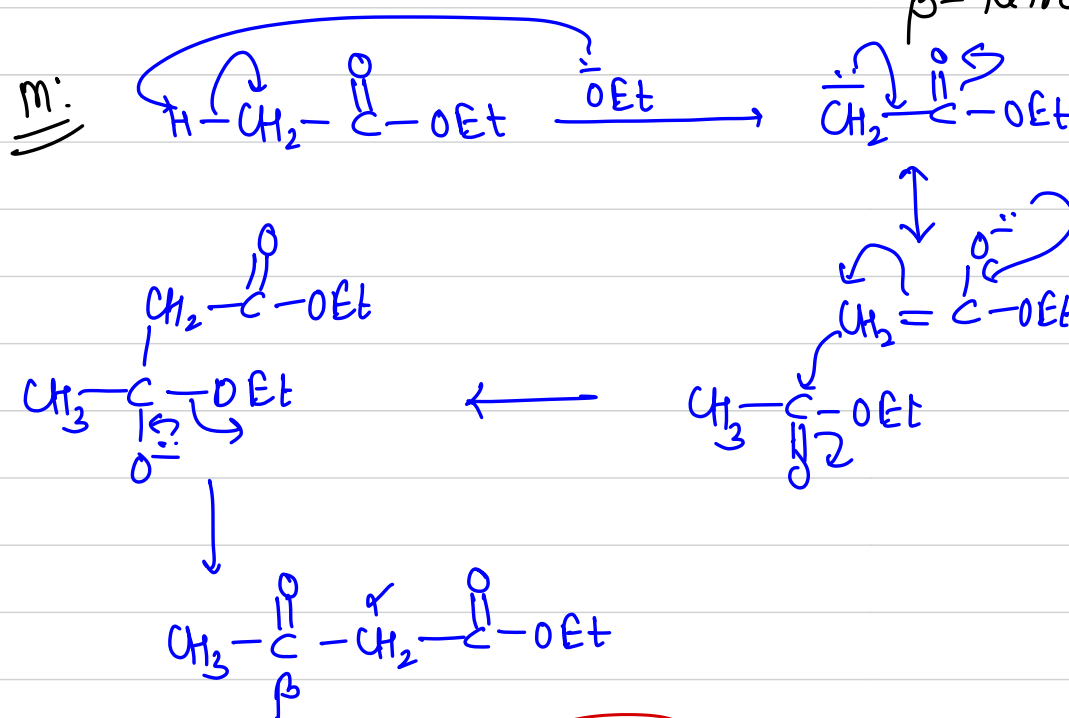
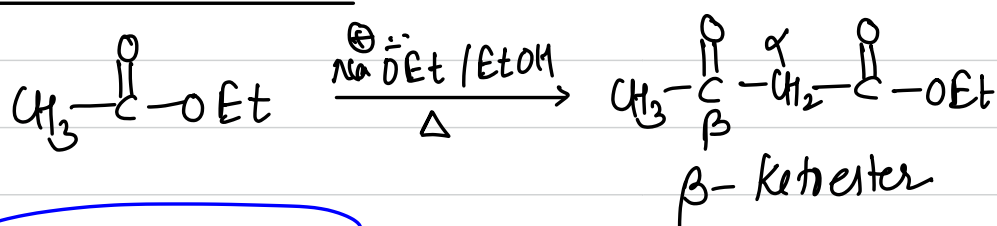
⊛ Aldol in acidic medium:-



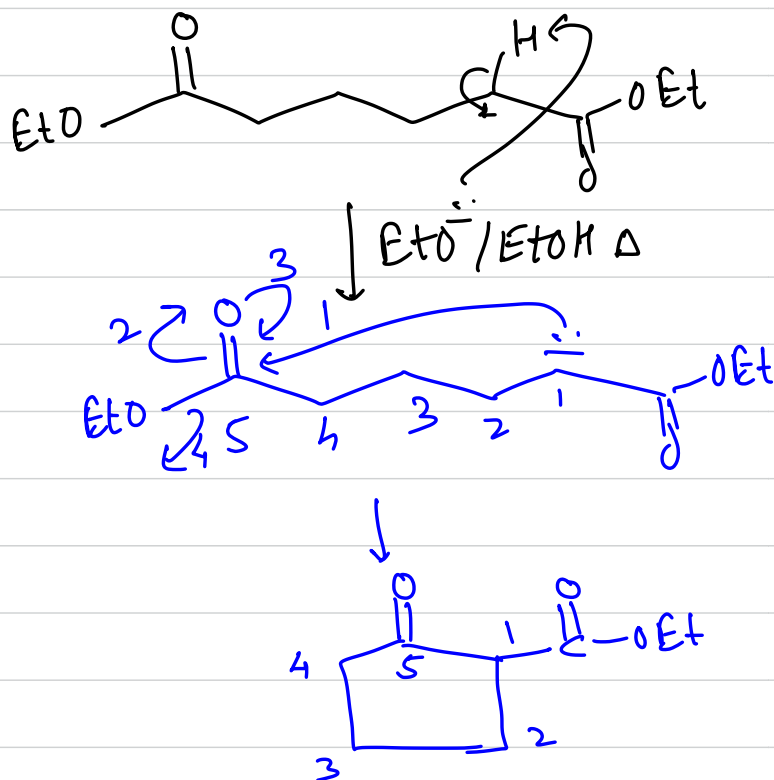
≡ m:



② Claisen Condensation :-



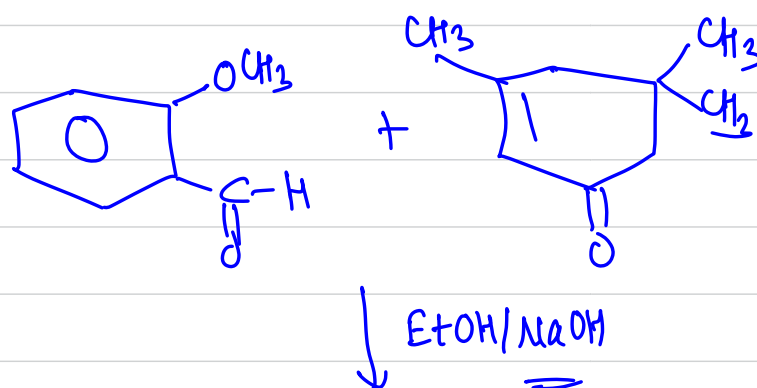
① Intramolecular claisen condensation :- (Diekmann cyclization)



③ Claisen-Schmidt Rxn:-

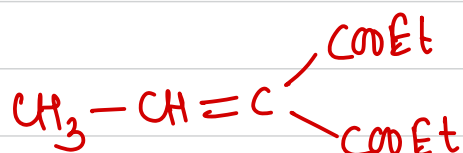
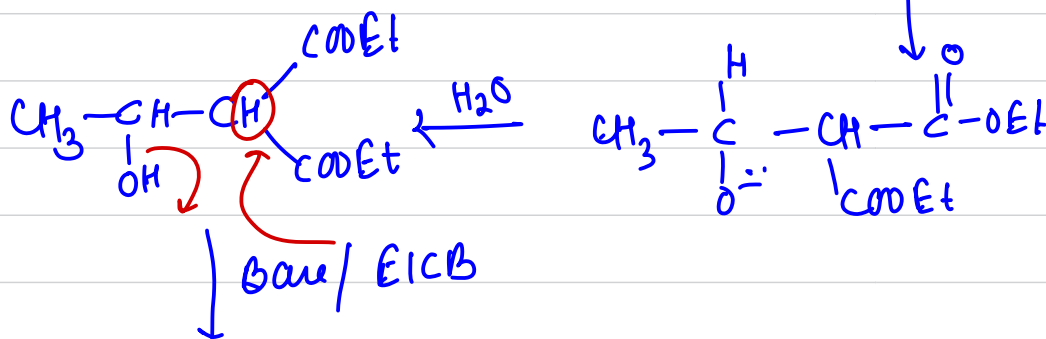
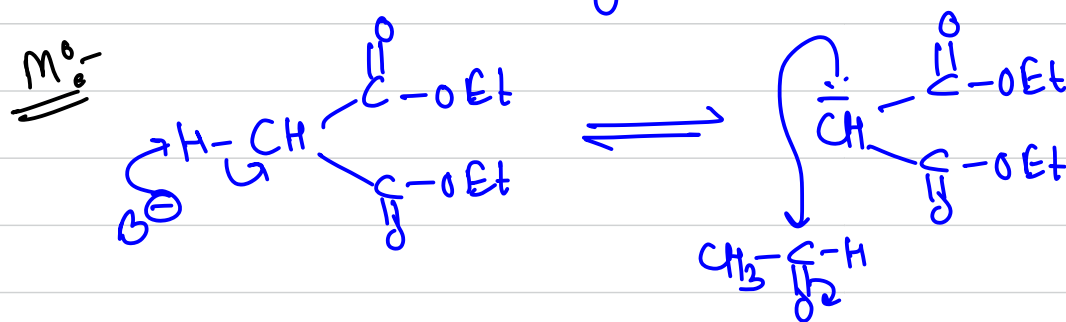
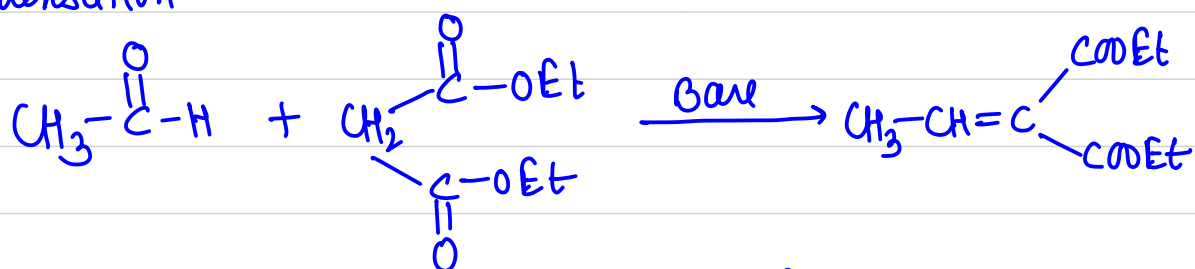
Successful cross aldol condensation is also known as claisen schmidt Rxn.

(hw)



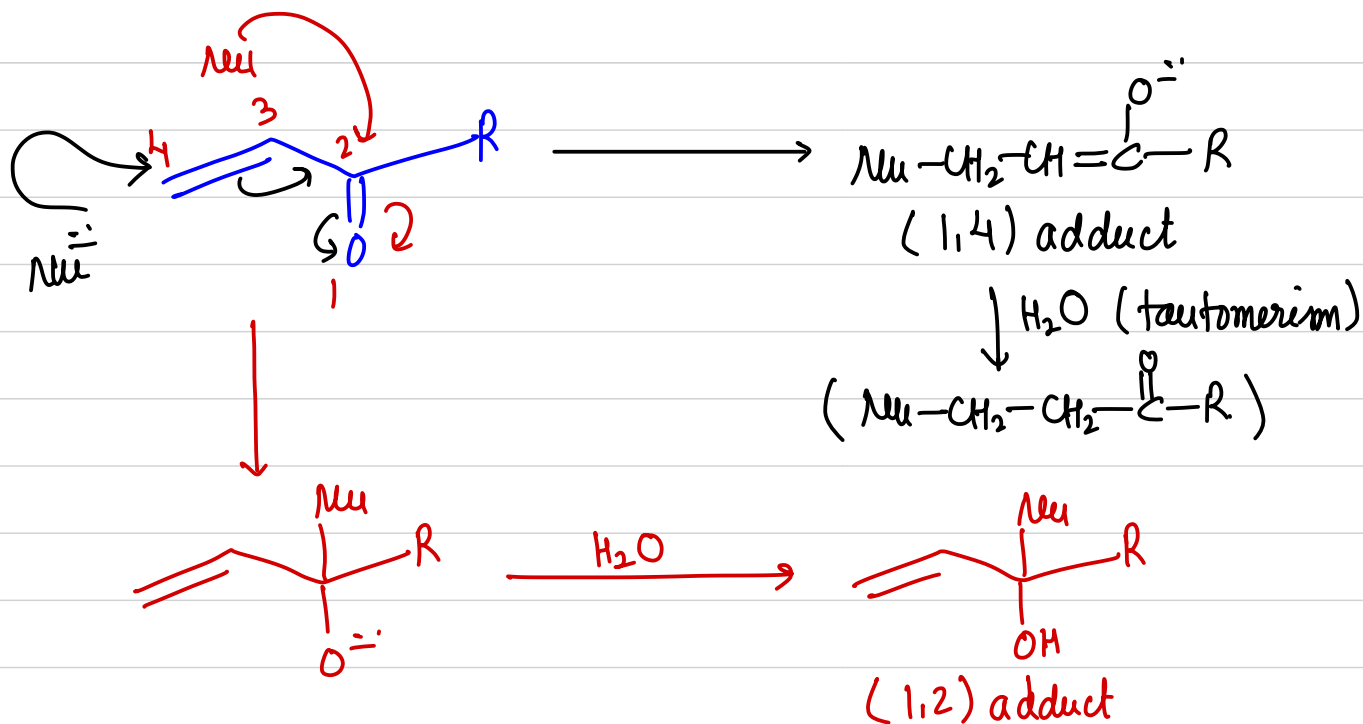
④ Knoevenagel Condensation^o

The condensation of aldehydes or ketones with active methylene compounds especially malonic esters in presence of very weak base like ammonia or amines is known as Knoevenagel Condensation.



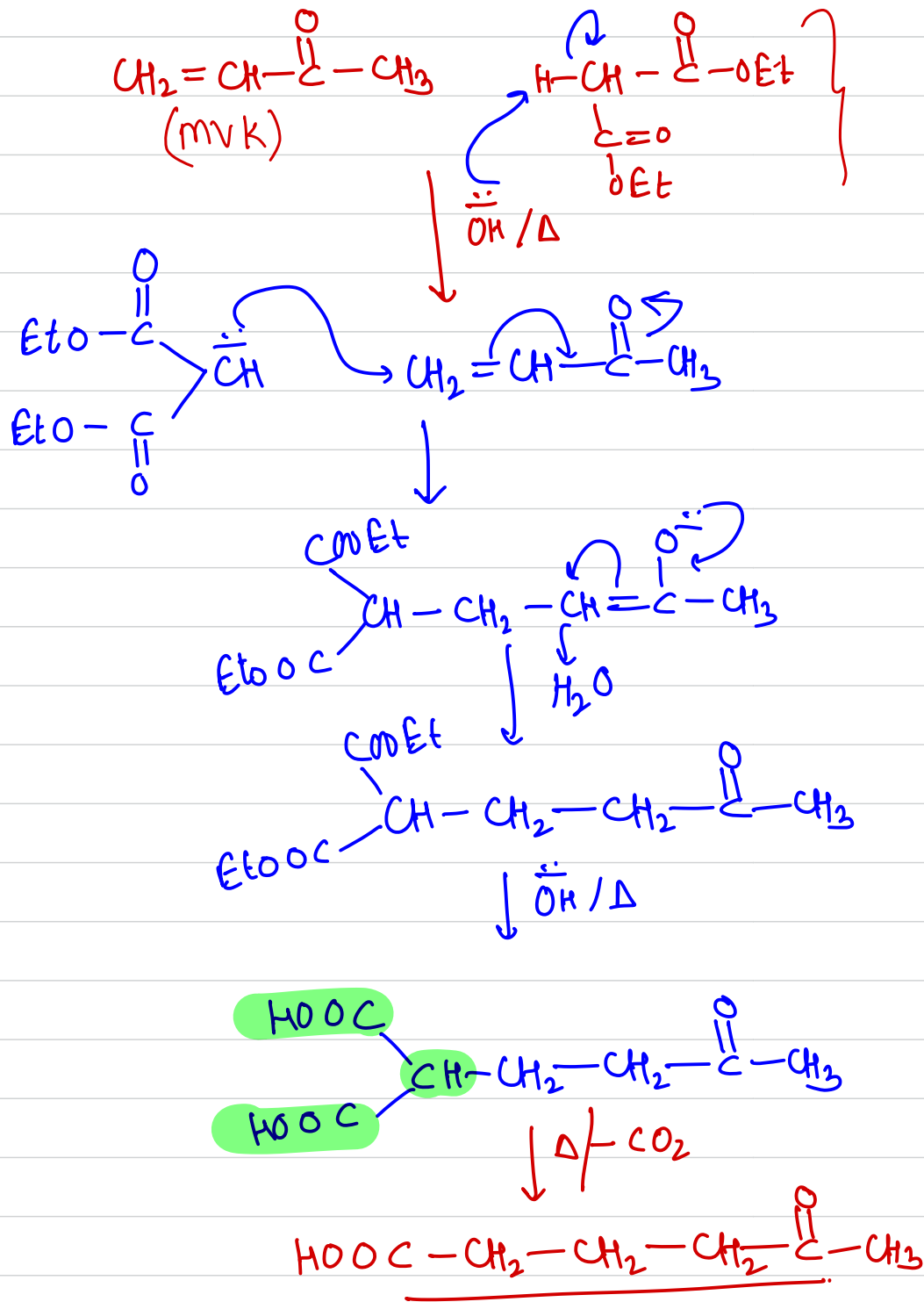
⑤ Michael Addition:- (Conjugate addition)

Addition of nucleophiles on α, β unsaturated aldehydes or ketones is known as Michael Addition.

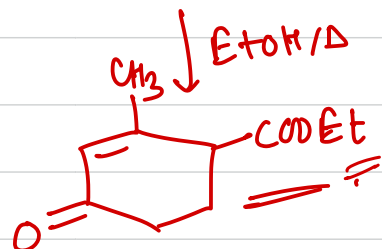
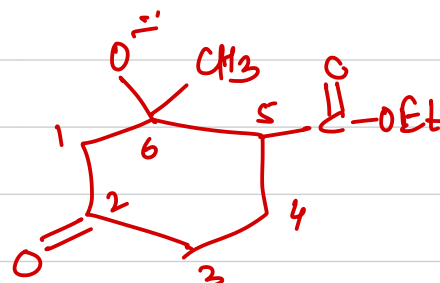
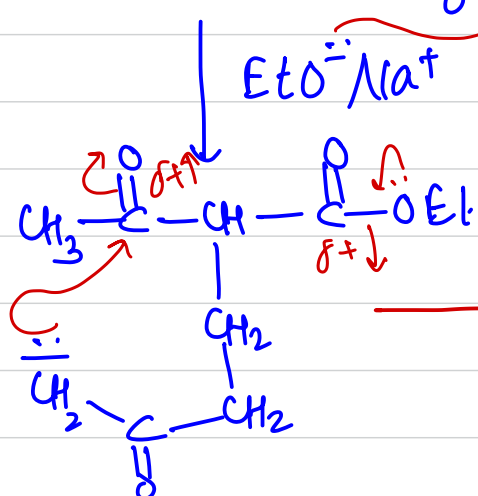
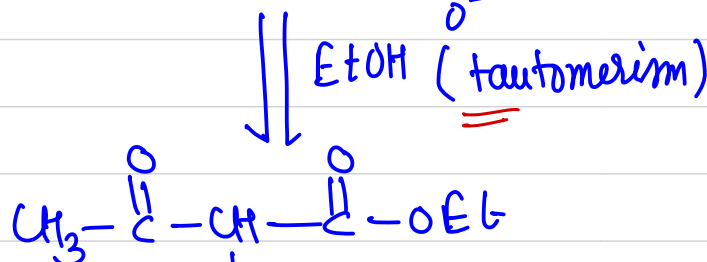
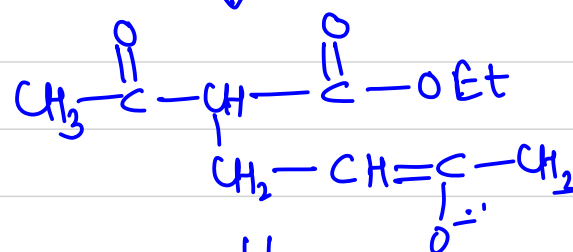
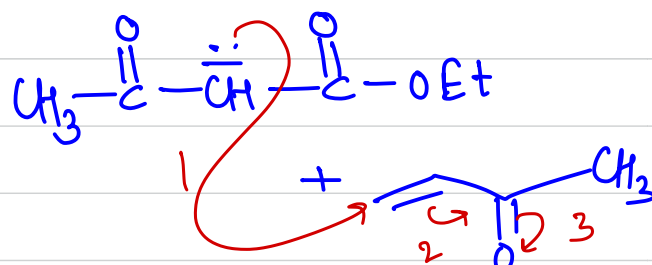
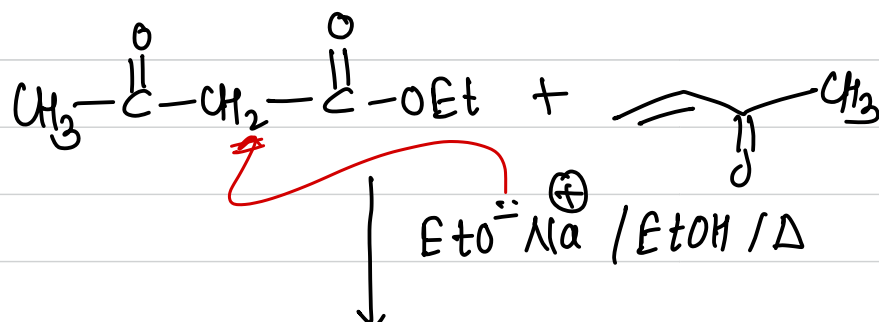


* As the weak nucleophiles are used in Michael Addition. therefore major product is formed by (1,4) adduct.

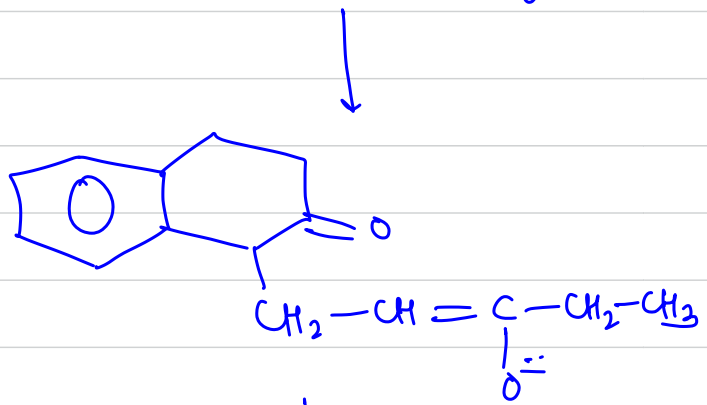
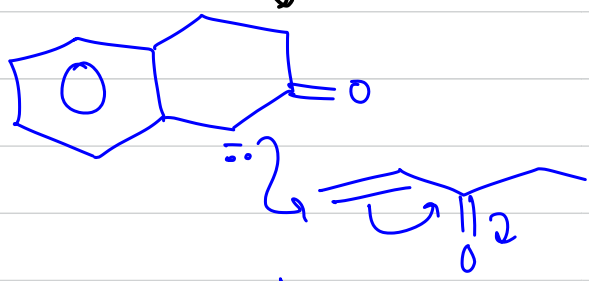
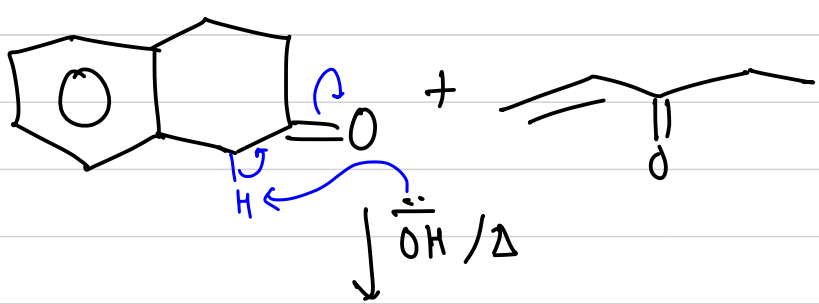
Michael donor	Michael acceptors
$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\ddot{\text{C}}\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R}$	$\text{CH}_2=\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R}$
$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\ddot{\text{C}}\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OR}$	$\text{CH}_2=\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
R_2CuLi	$\text{CH}_2=\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OR}$
$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\ddot{\text{C}}\text{H}-\text{C}\equiv\text{N}$	$\text{CH}_2=\text{CH}-\text{C}\equiv\text{N}$
$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\ddot{\text{C}}\text{H}-\text{N}=\text{O}$	



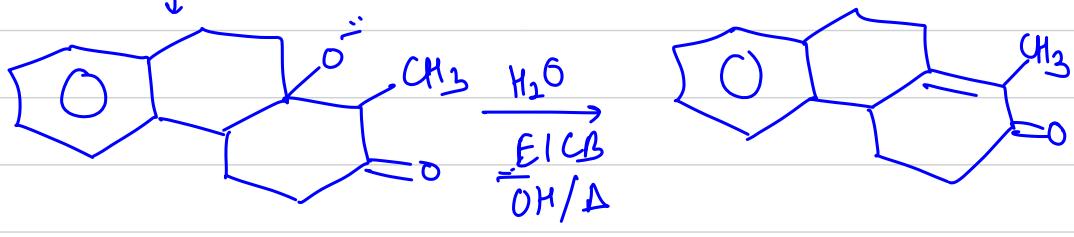
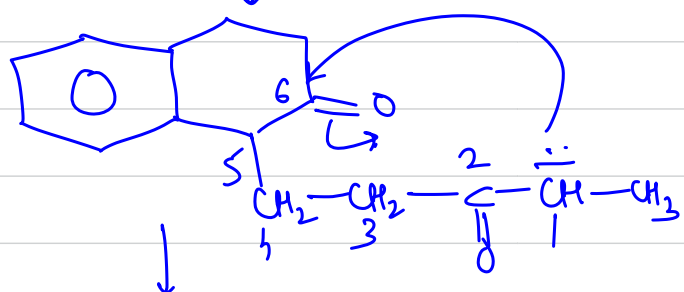
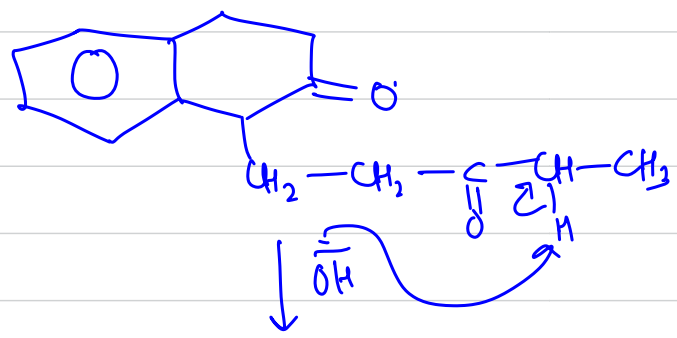
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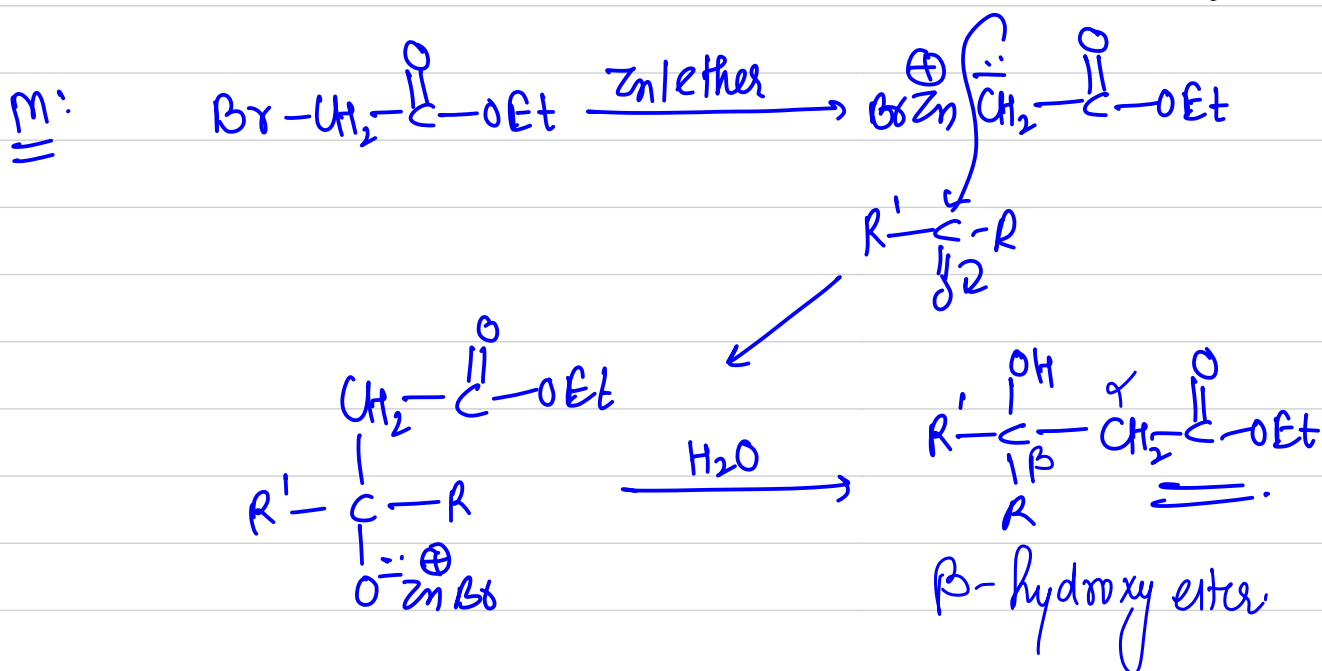
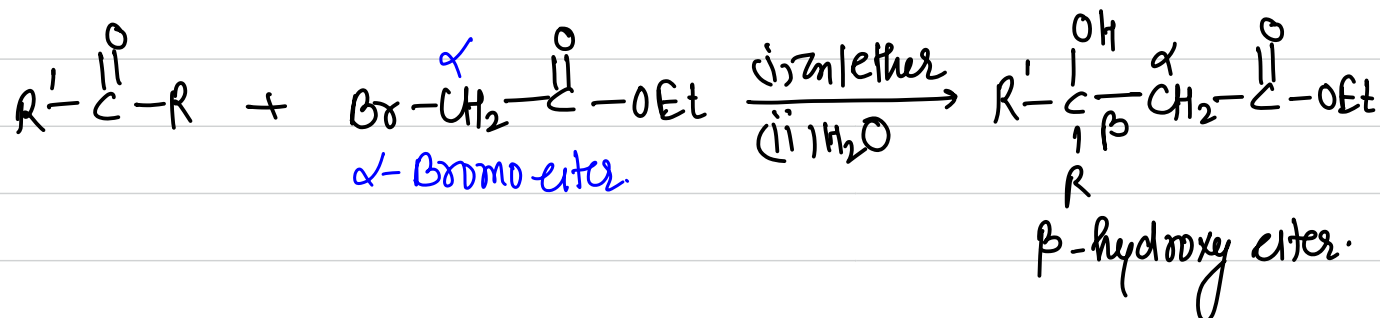


$\downarrow \text{H}_2\text{O (tautomerism)}$

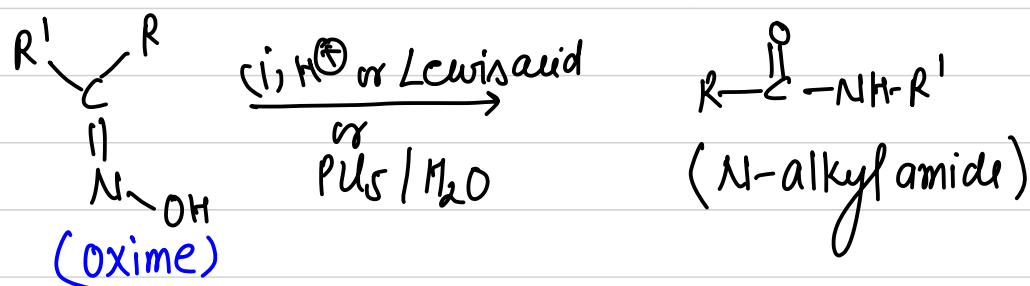


⑥ Reformatsky reaction:-

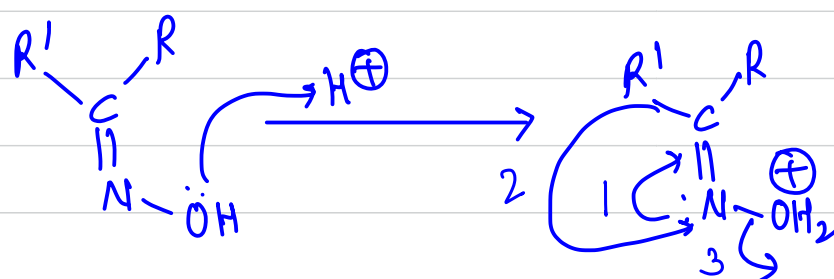
The reaction of α -bromoester with carbonyl compounds to form β -hydroxy ester in presence of Zn & inert solvent is called reformatsky rxn.



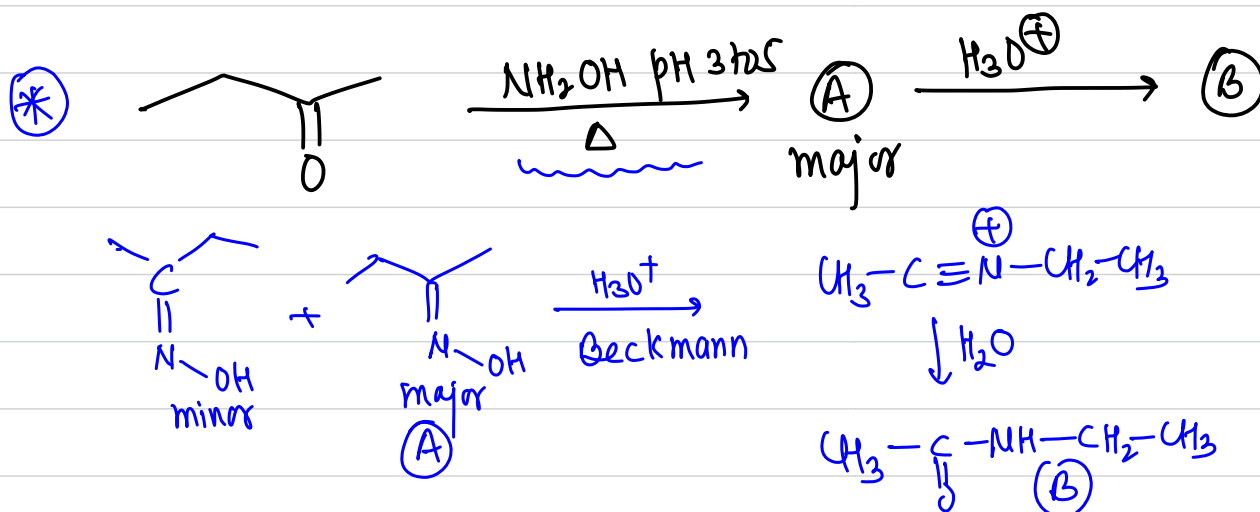
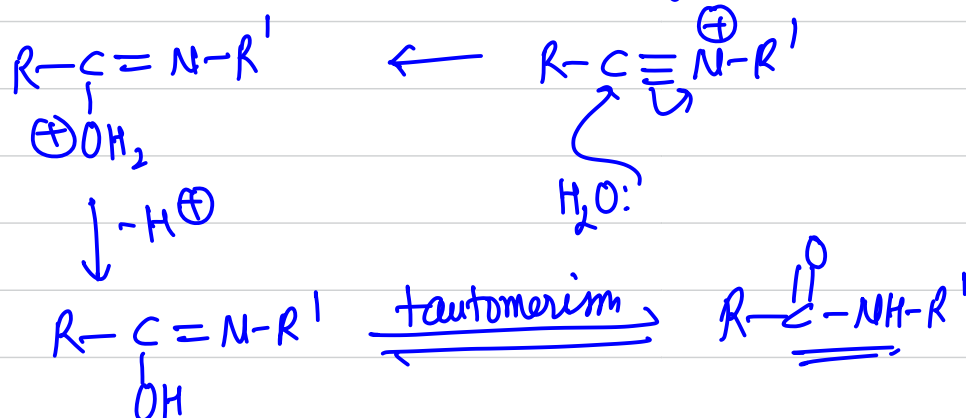
⑦ Beckmann Rearrangement:-



m:



that group will migrate which is anti to -OH.

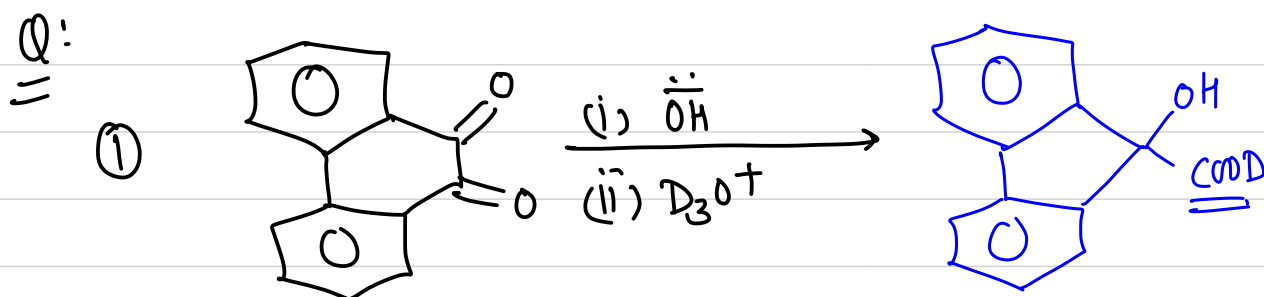
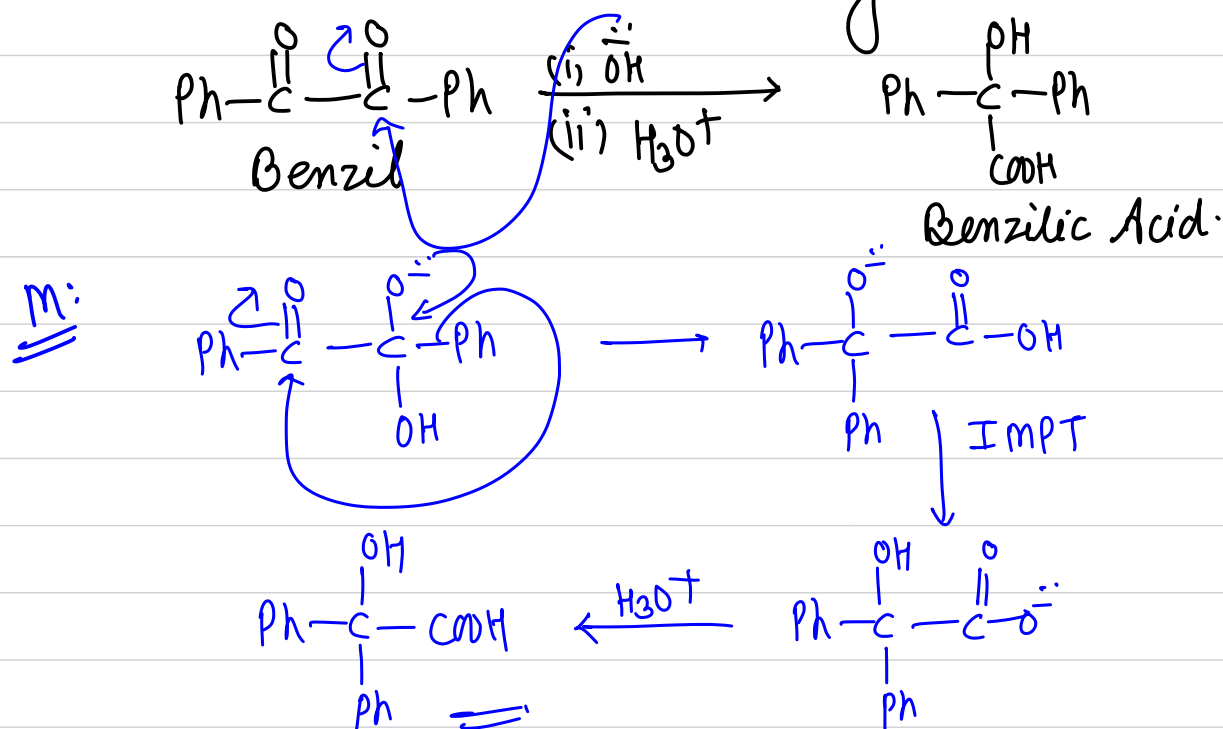


Q-2 An organic compound having molecular formula $C_6H_{10}O$.
(A). on rxn with hydroxyl amine in presence of heat & pH 3 to 5 gives (B). (B) on rxn with H_2O^+ gives (C).

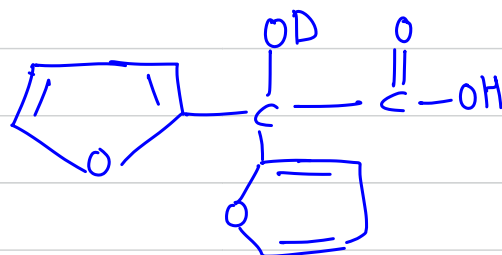
(C) on hydrolysis gives (D). (D) on condensation gives (E) which is a polymer.

what is name of polymer & also find out (A) to (E)

⑧ Benzil-Benzilic Acid Rearrangement

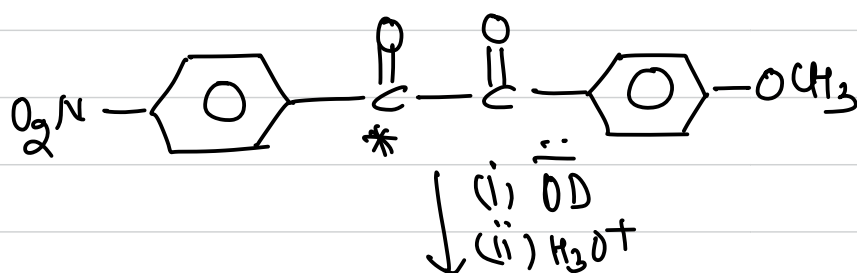






HW

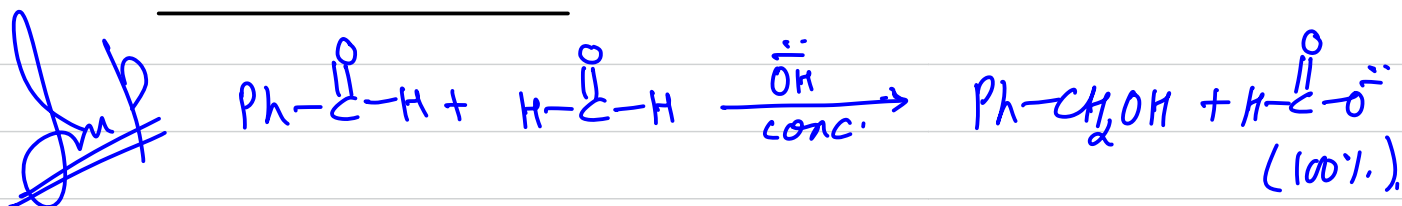
③



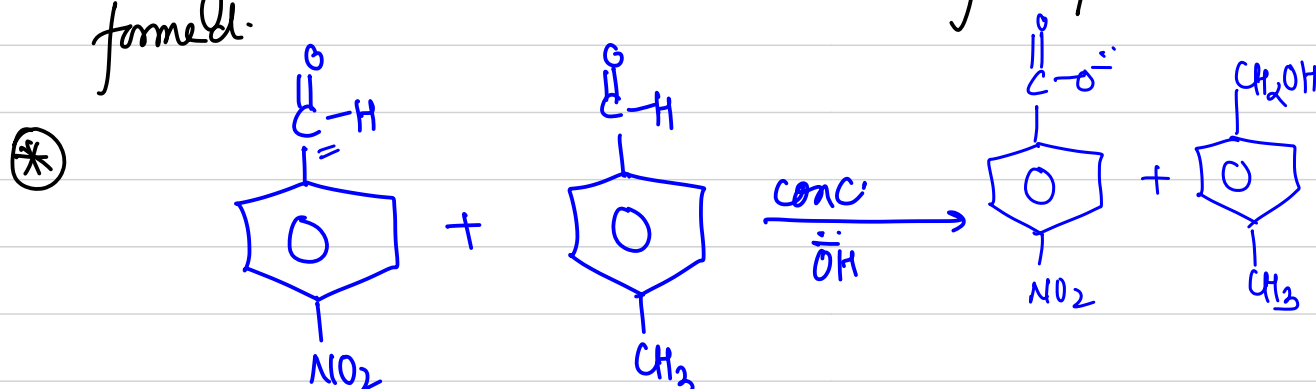
attack will be that carbon which is having more δ^+ charge.



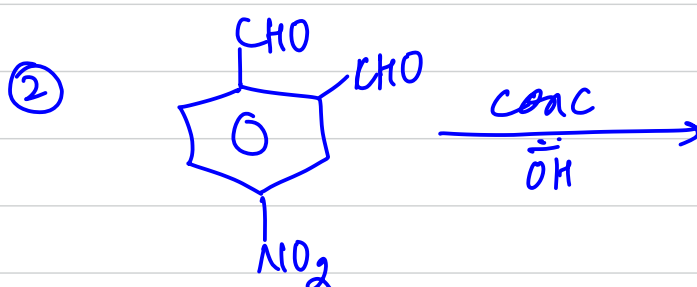
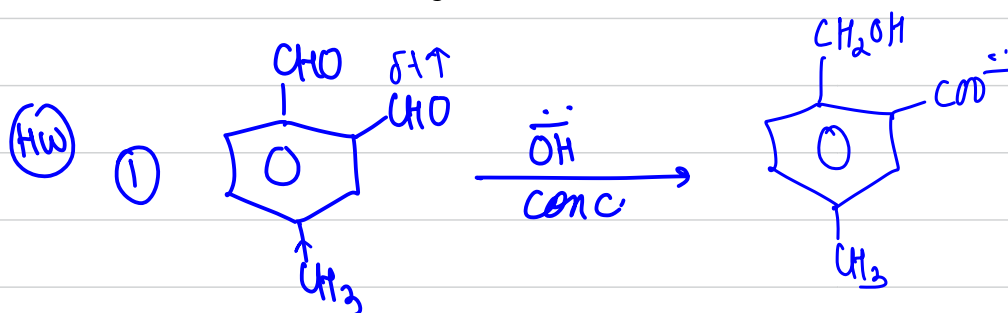
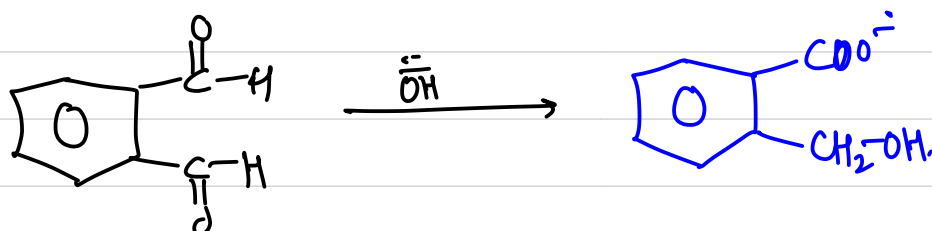
Cross Cannizaro:-



* In general in cross Cannizaro rxn four products are formed.

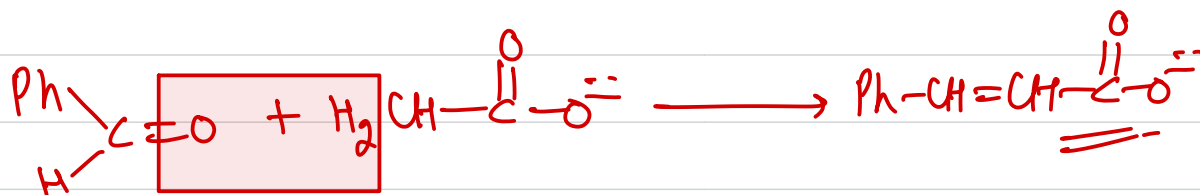
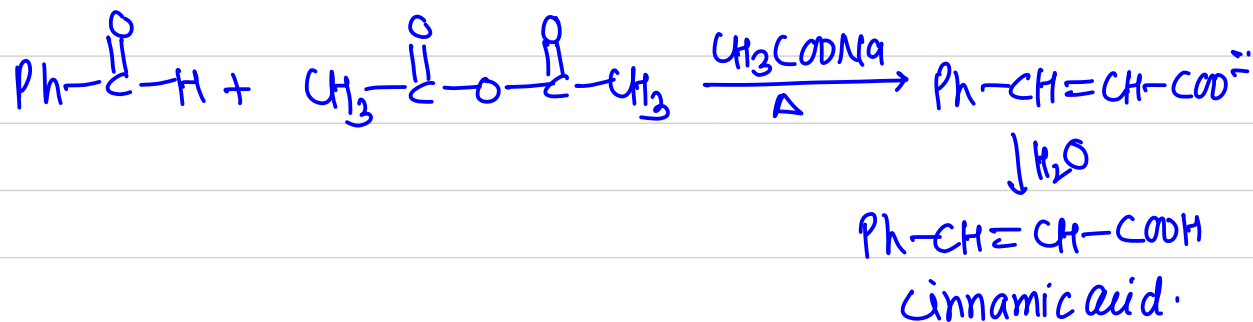


* Intra molecular Cannizaro:-



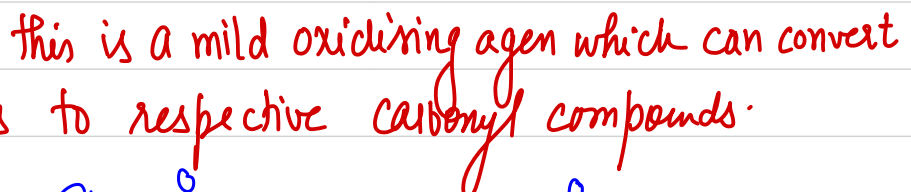
⑩ Imp Perkin Reaction:-

The reaction of aromatic aldehydes with acid anhydride in presence of sodium salt of the acid is called perkin's Rxn.

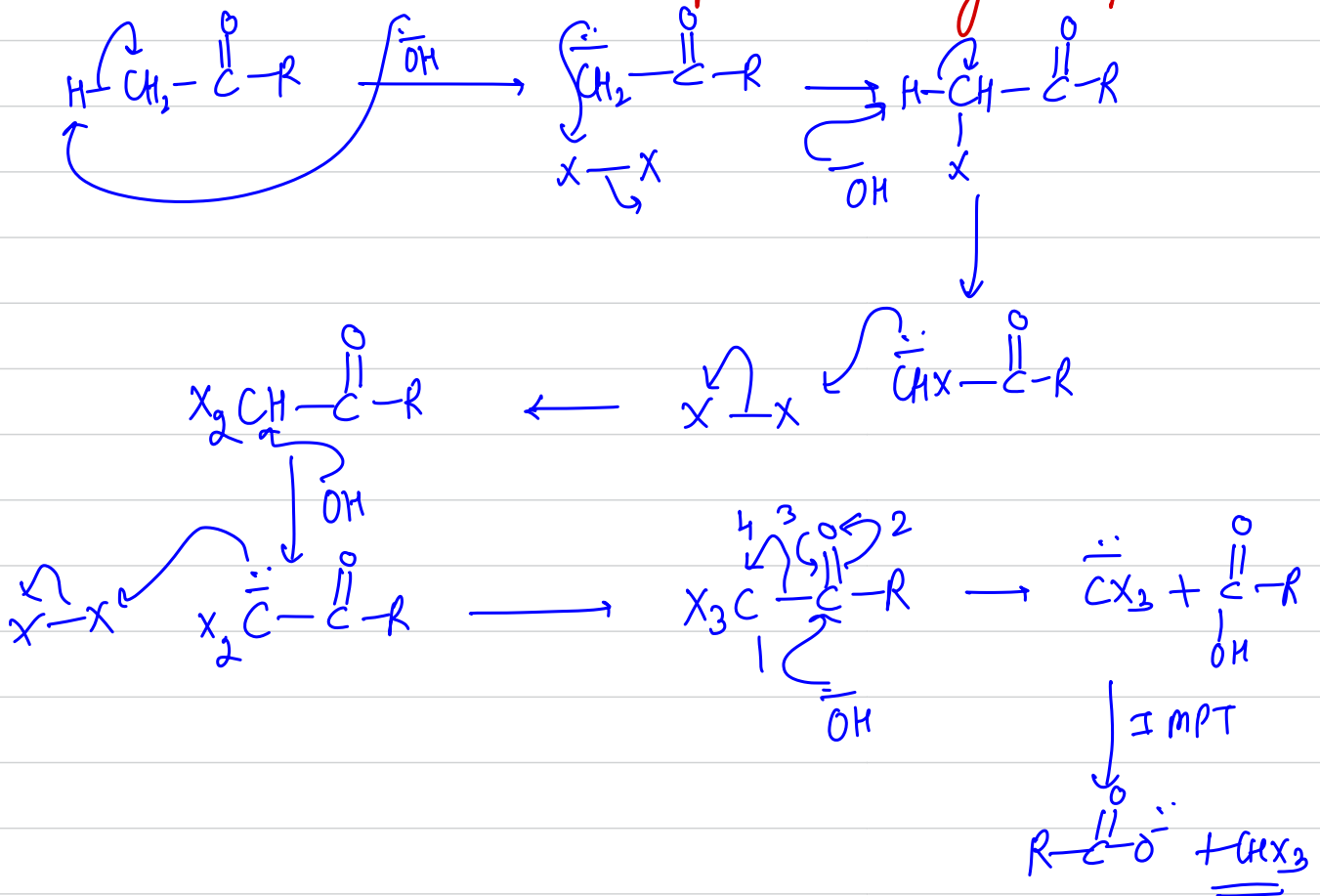


Reaction of methyl ketones or acetaldehyde in presence of X_2/OH^- to form haloform is called haloform rxn.

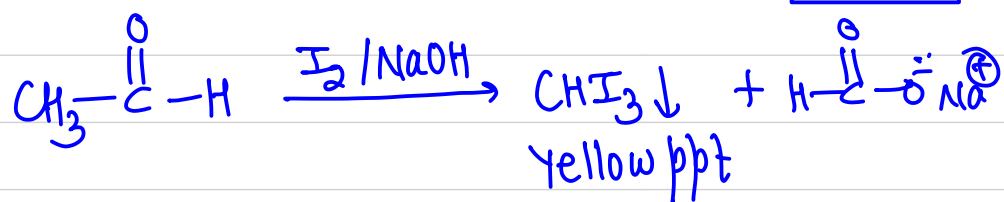
$\text{CHI}_3 \rightarrow \text{Iodoform} \rightarrow \text{Yellow solid (ppt).}$



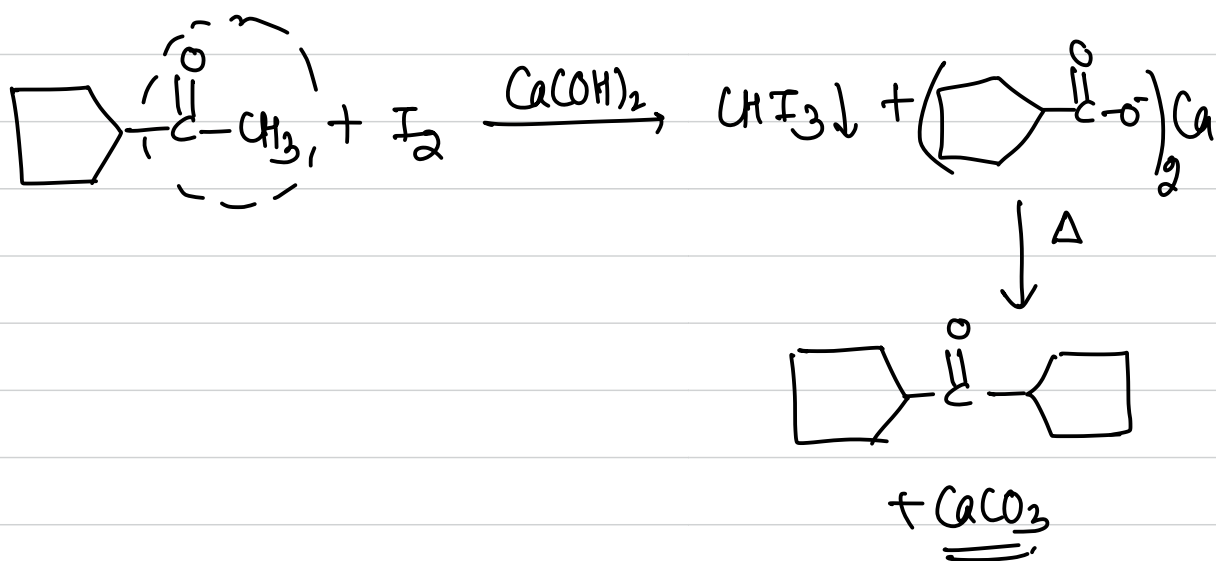
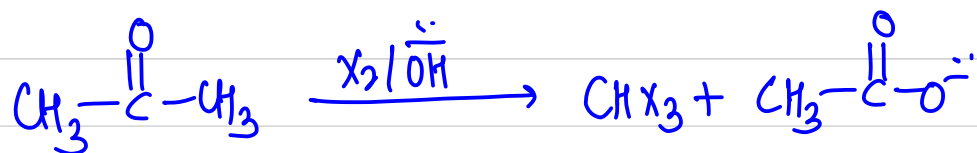
m:



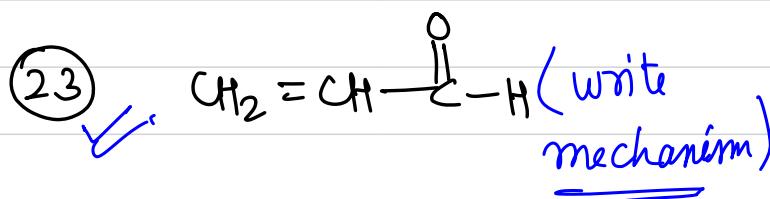
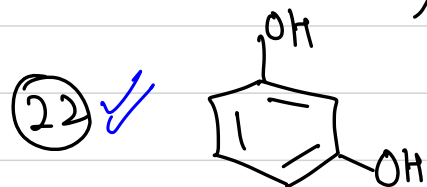
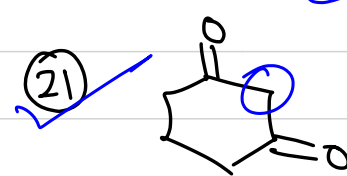
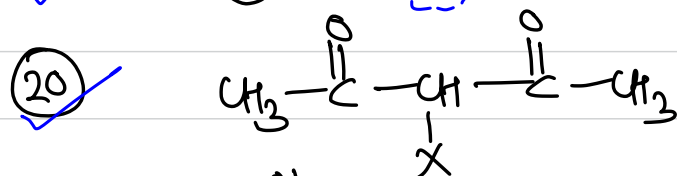
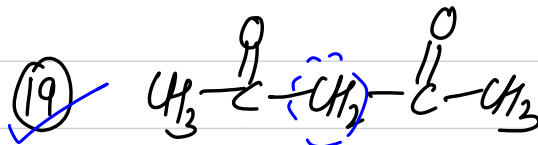
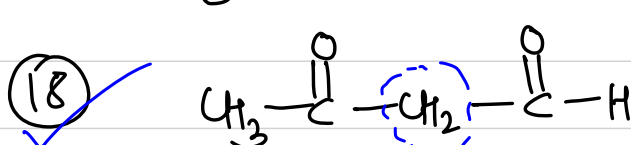
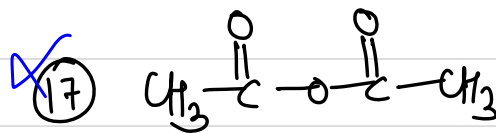
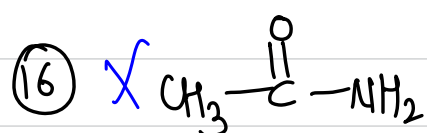
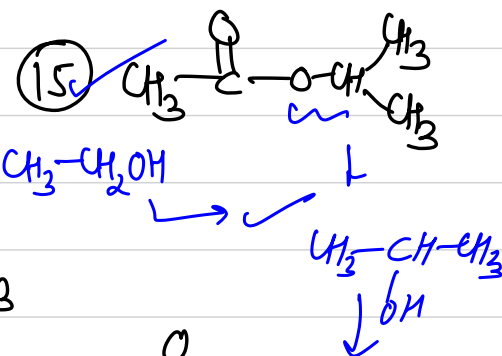
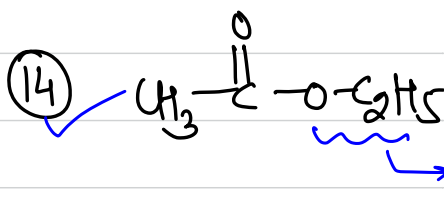
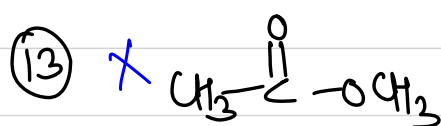
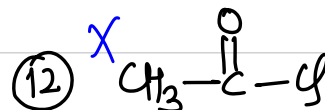
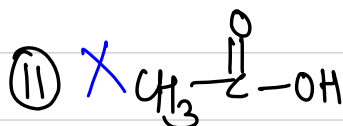
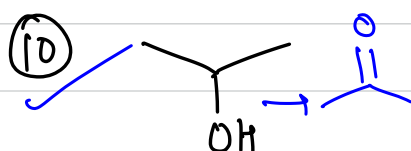
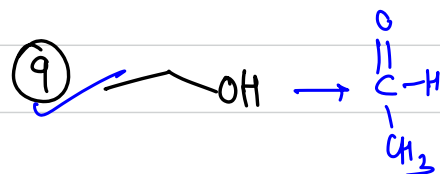
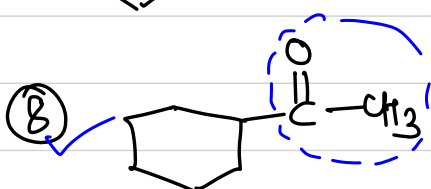
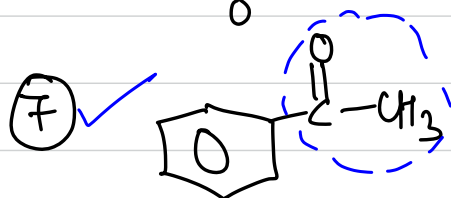
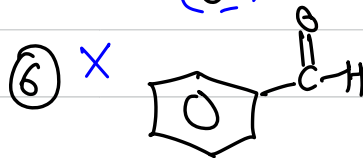
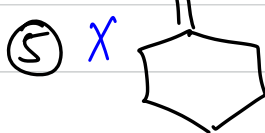
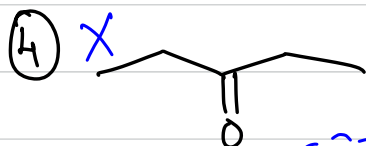
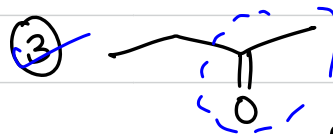
Ex!



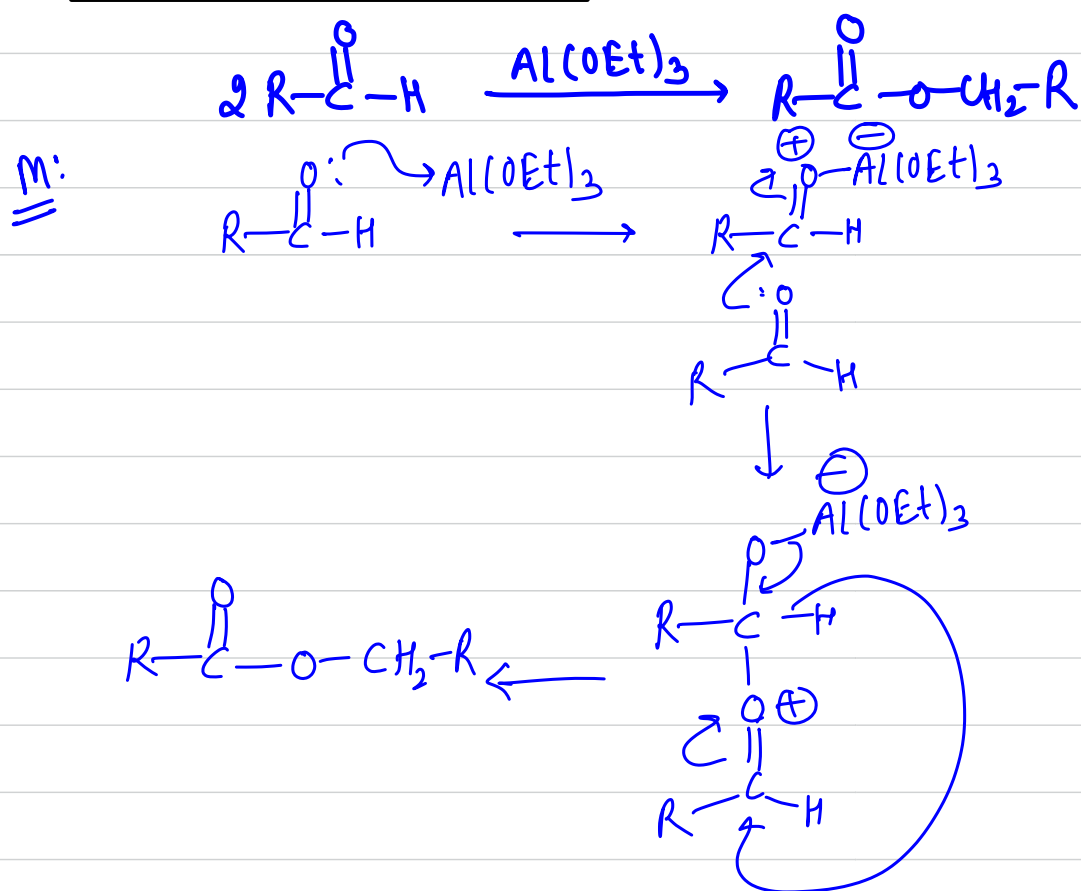
HCOONa



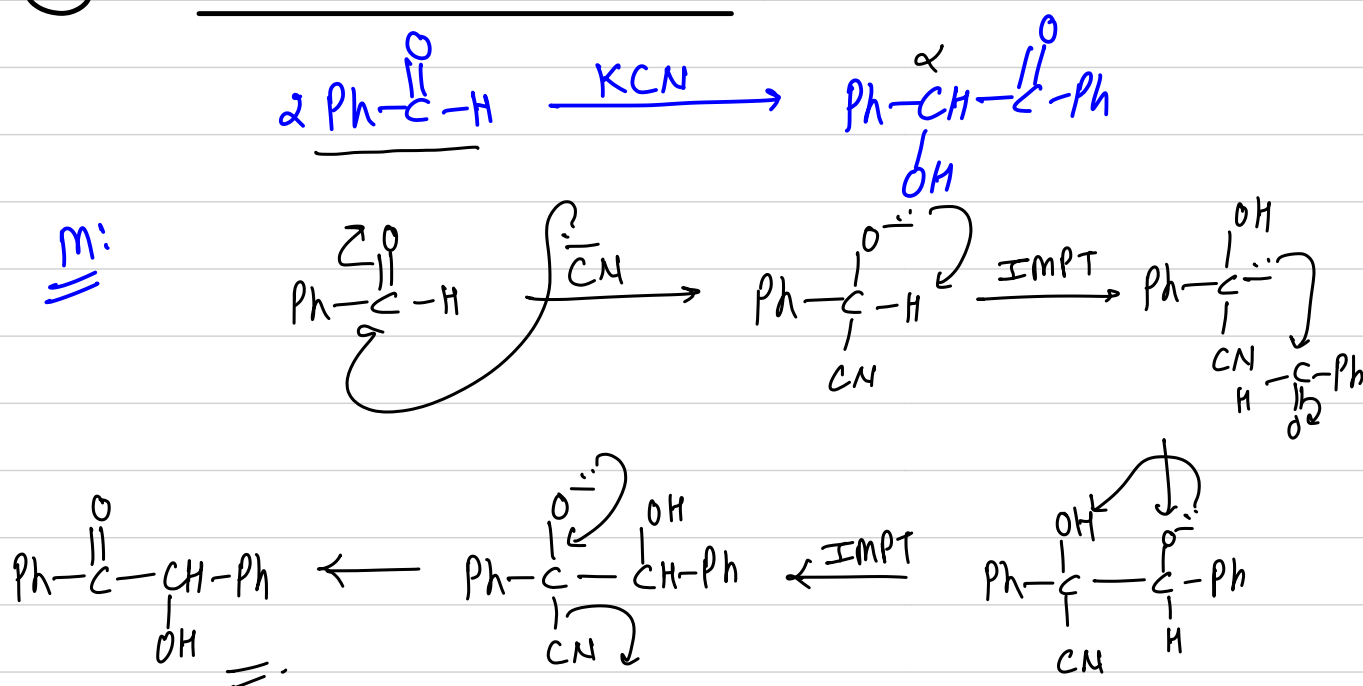
Q: Which of the following will give positive iodoform test?



⑫ Tischenko Reaction:-

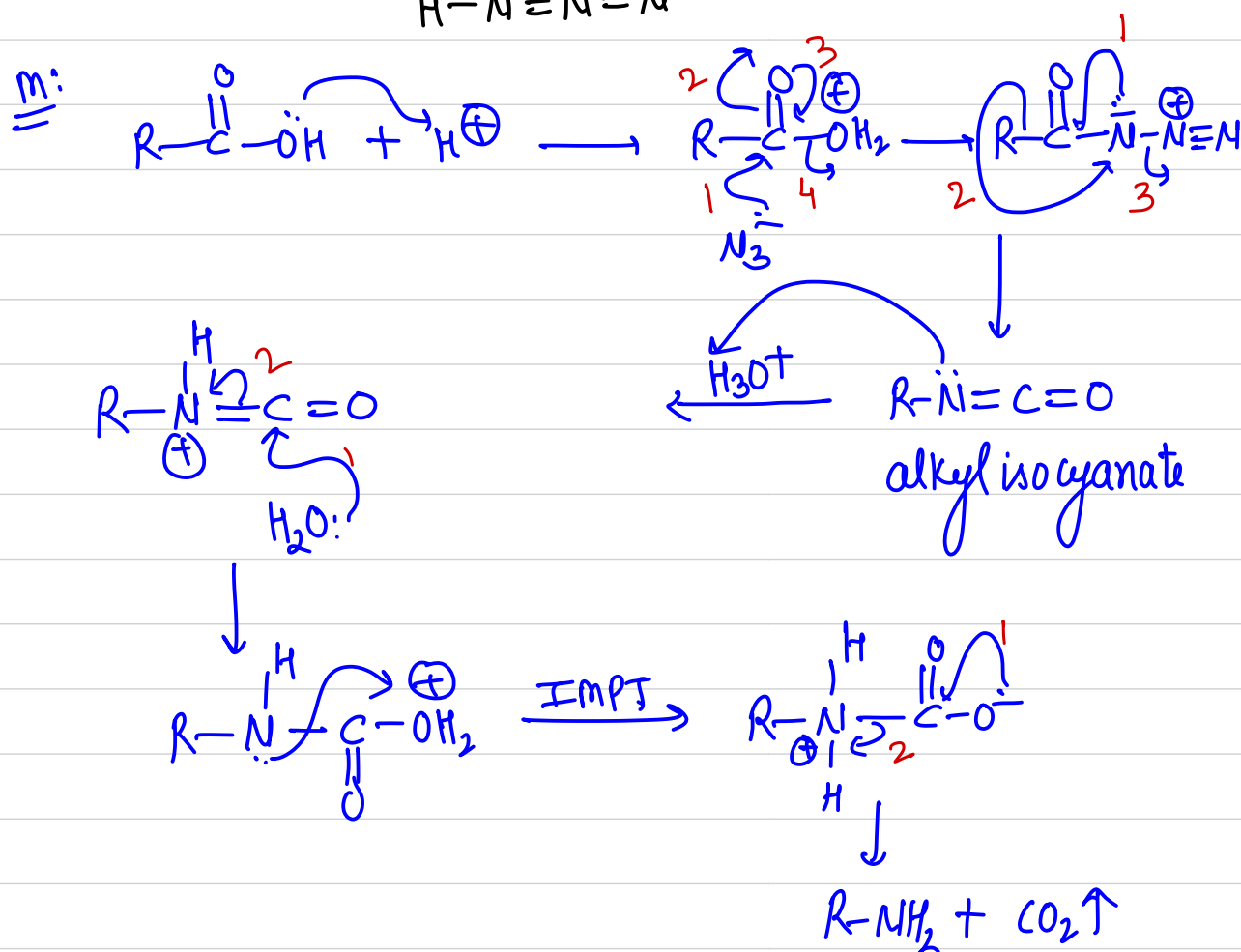
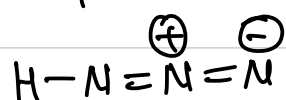
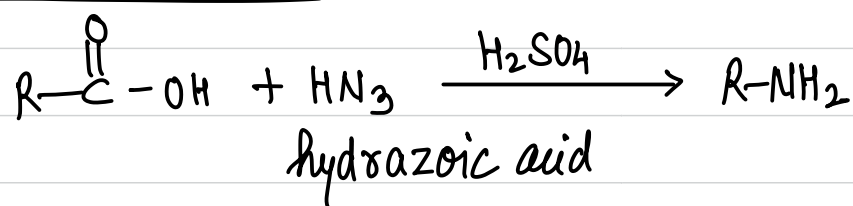


⑬ Benzoin Condensation:-



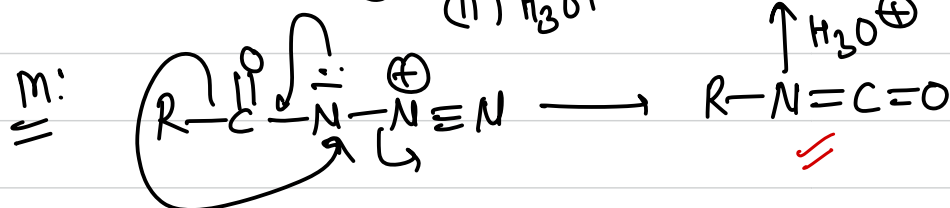
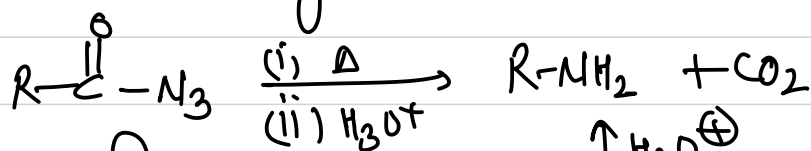
⑭

Schmidt Reaction:-



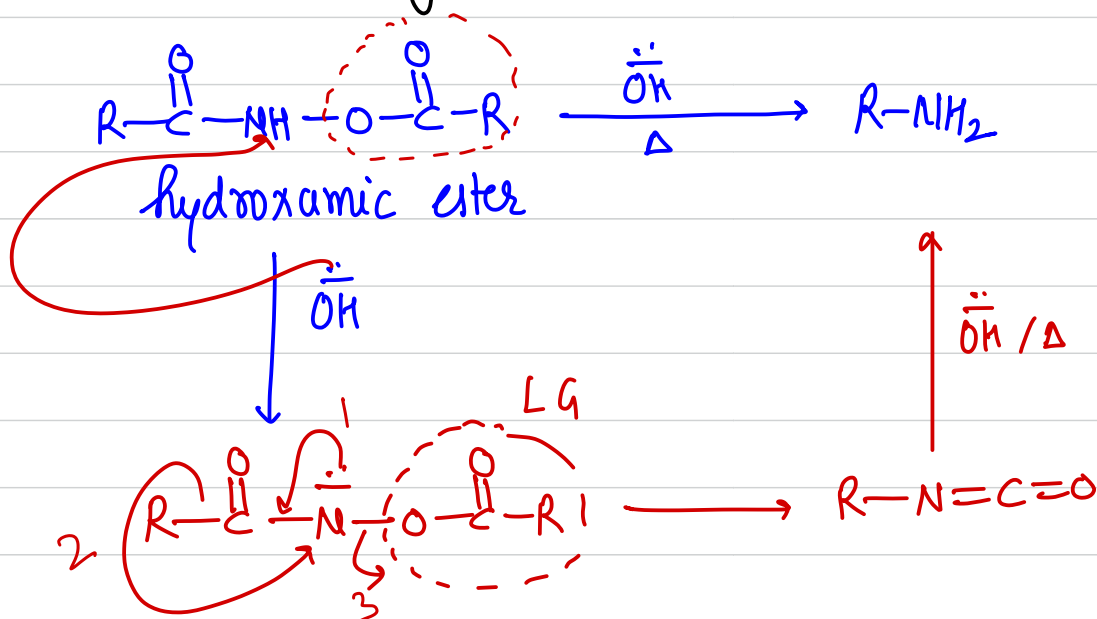
⑮

Curtius Rearrangement:-



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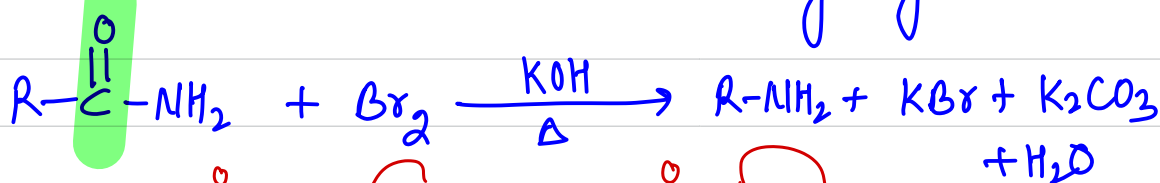
Lossen Rearrangement:-



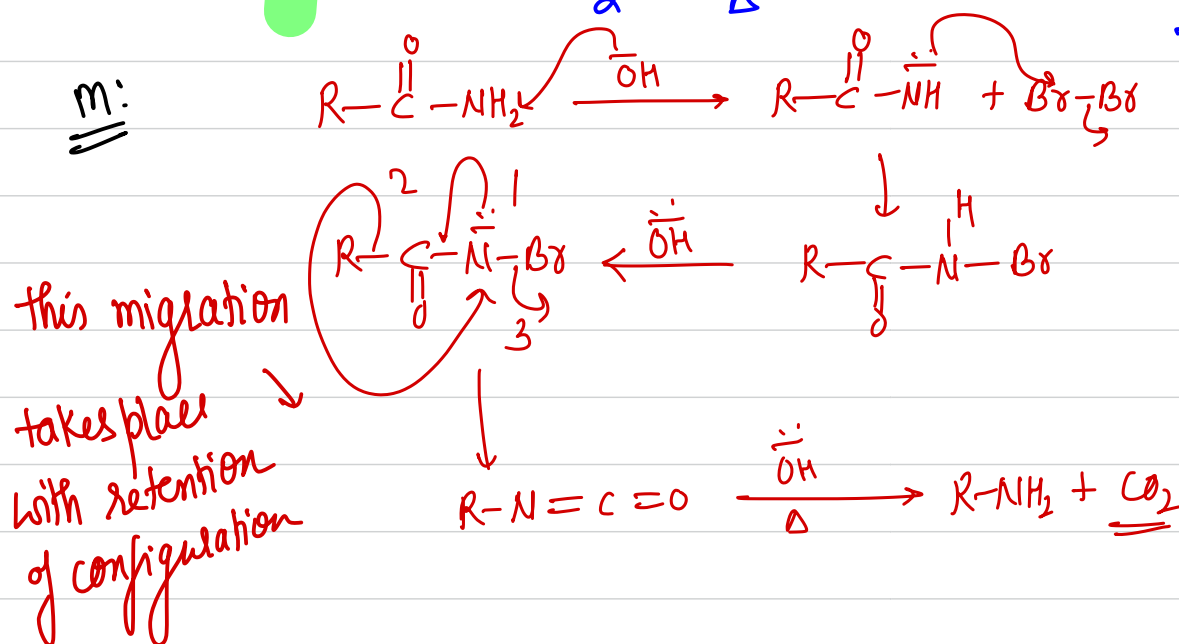
~~17~~

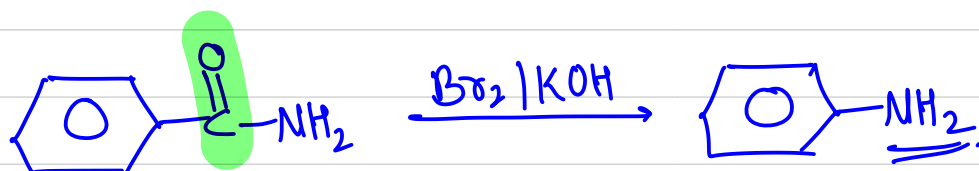
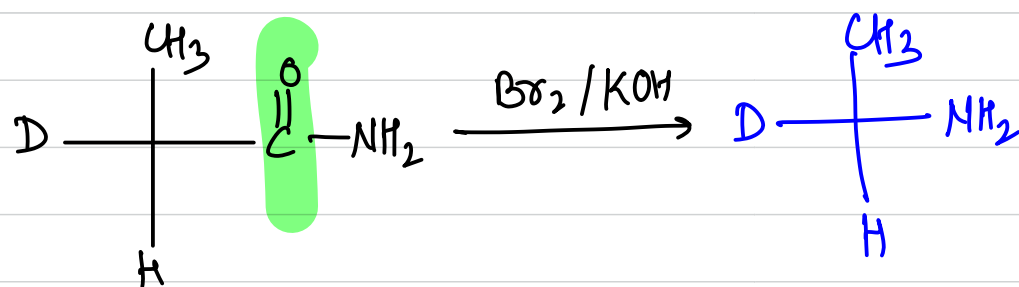
Hoffmann Bromamide Degradation:-

decrease in length of carbon chain.

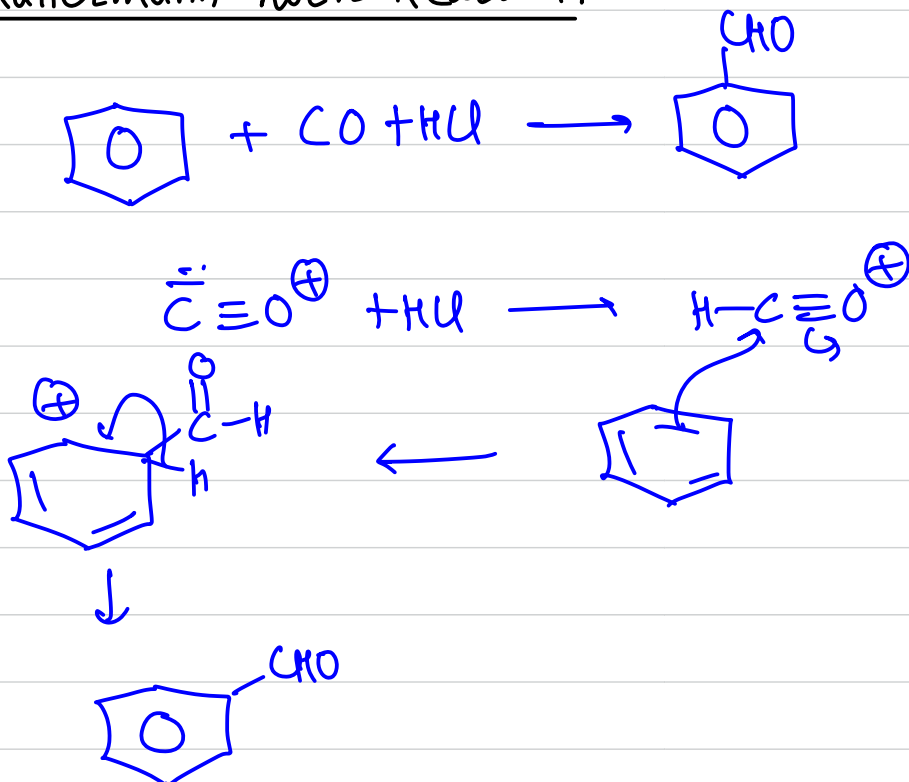


m:

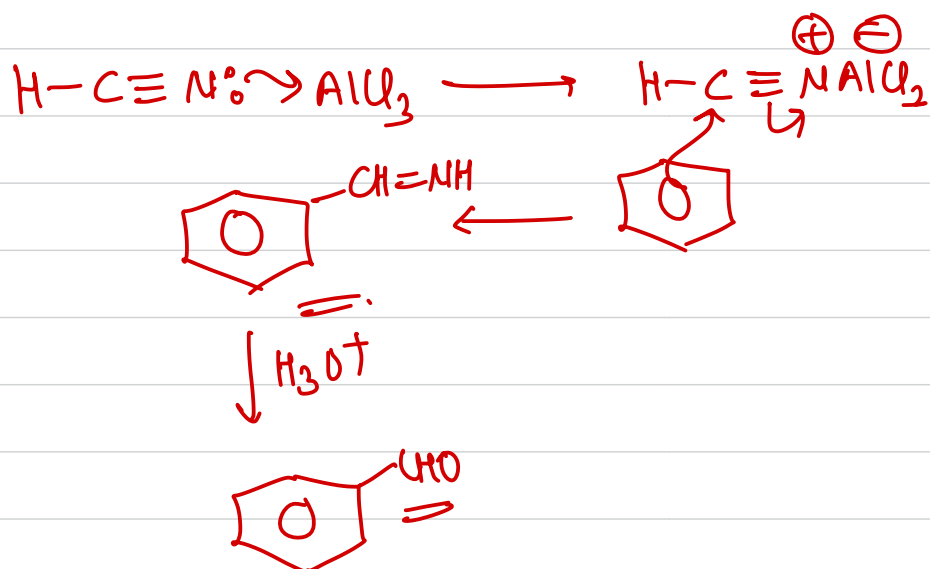
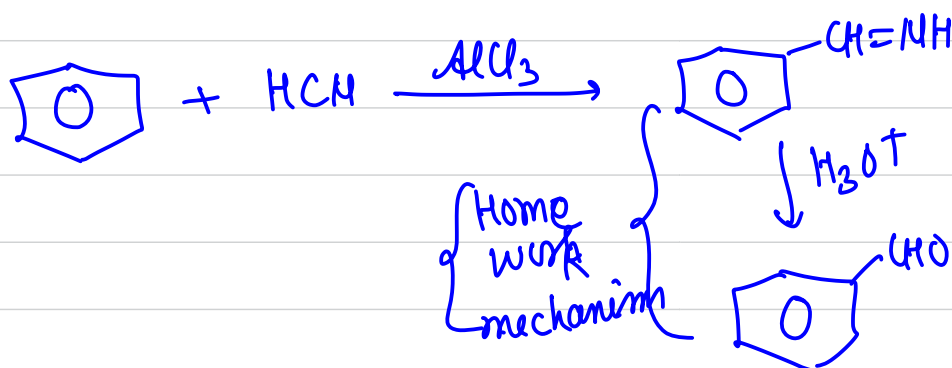




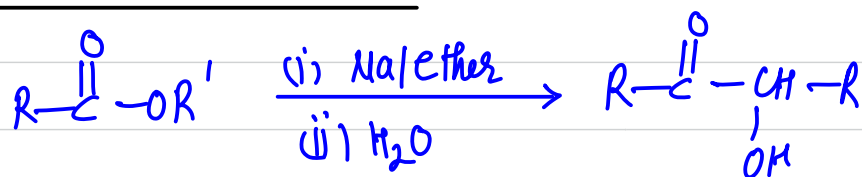
⑮ Gattermann Koch Reaction:-



⑰ Gattermann Reaction:-

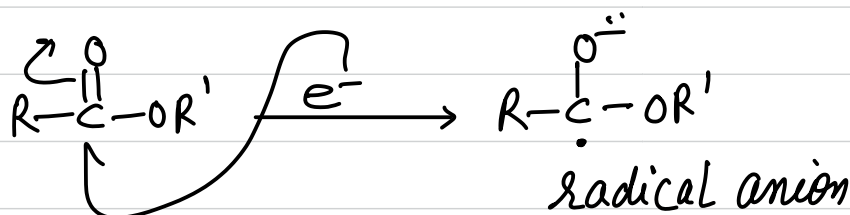


② Acyloin Condensation:-

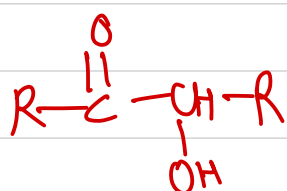
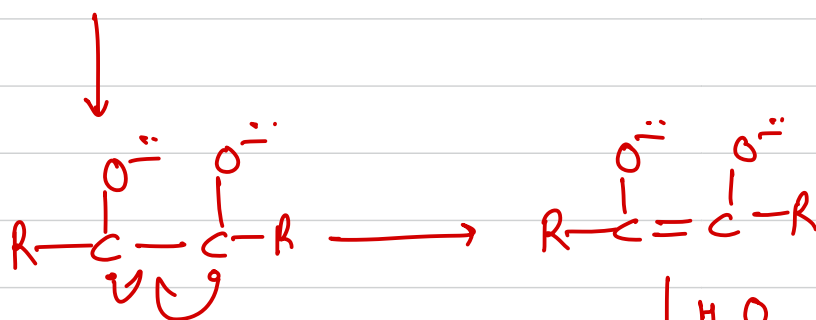
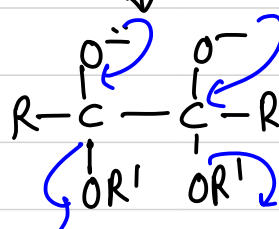
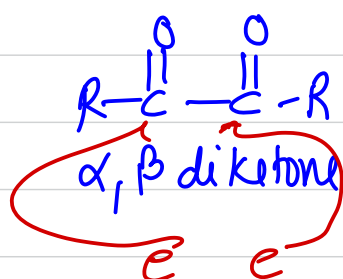


α -hydroxy carbonyls.

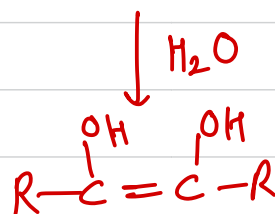
Mechanism:-



dimerization



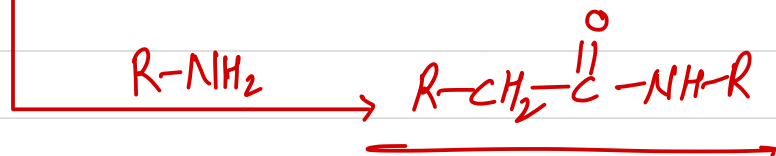
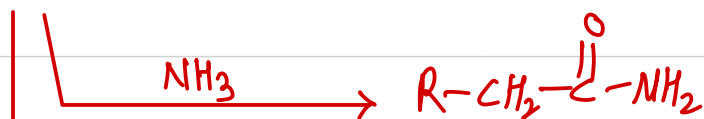
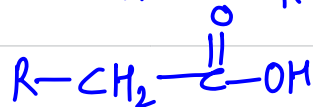
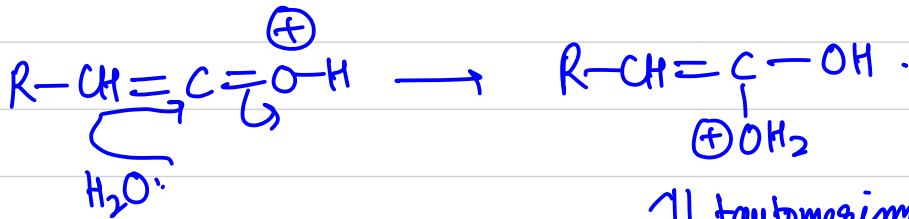
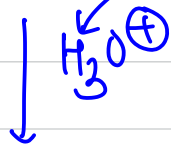
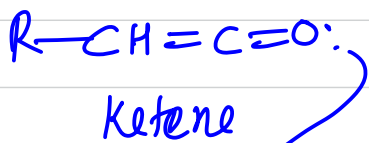
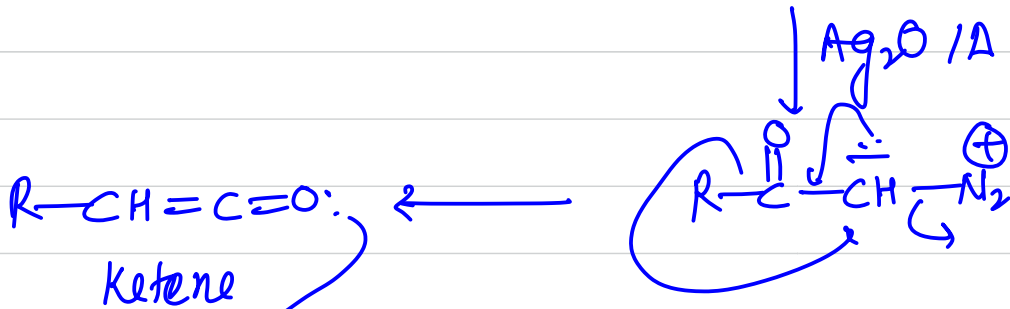
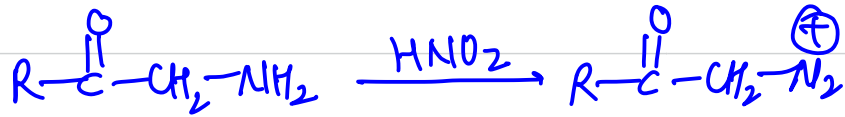
tautomerism



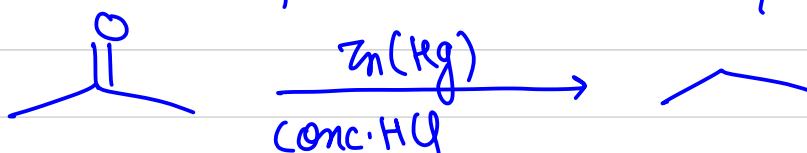
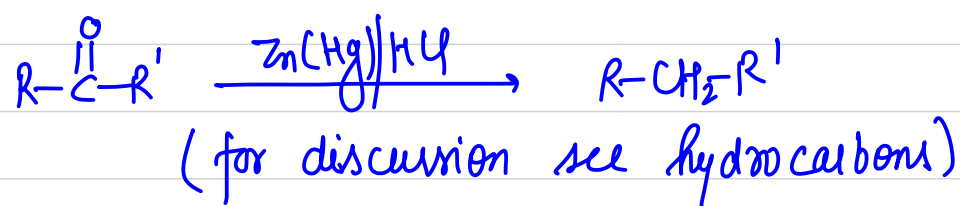
Final Product

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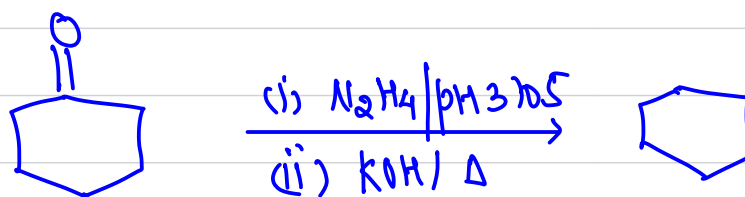
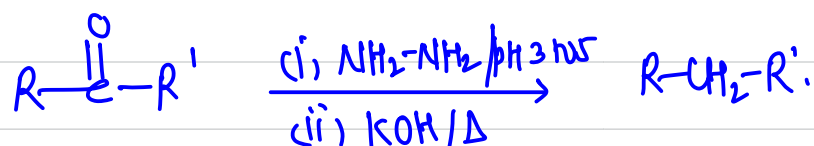
Witt Rearrangement



22) Clemenson's Reduction :-

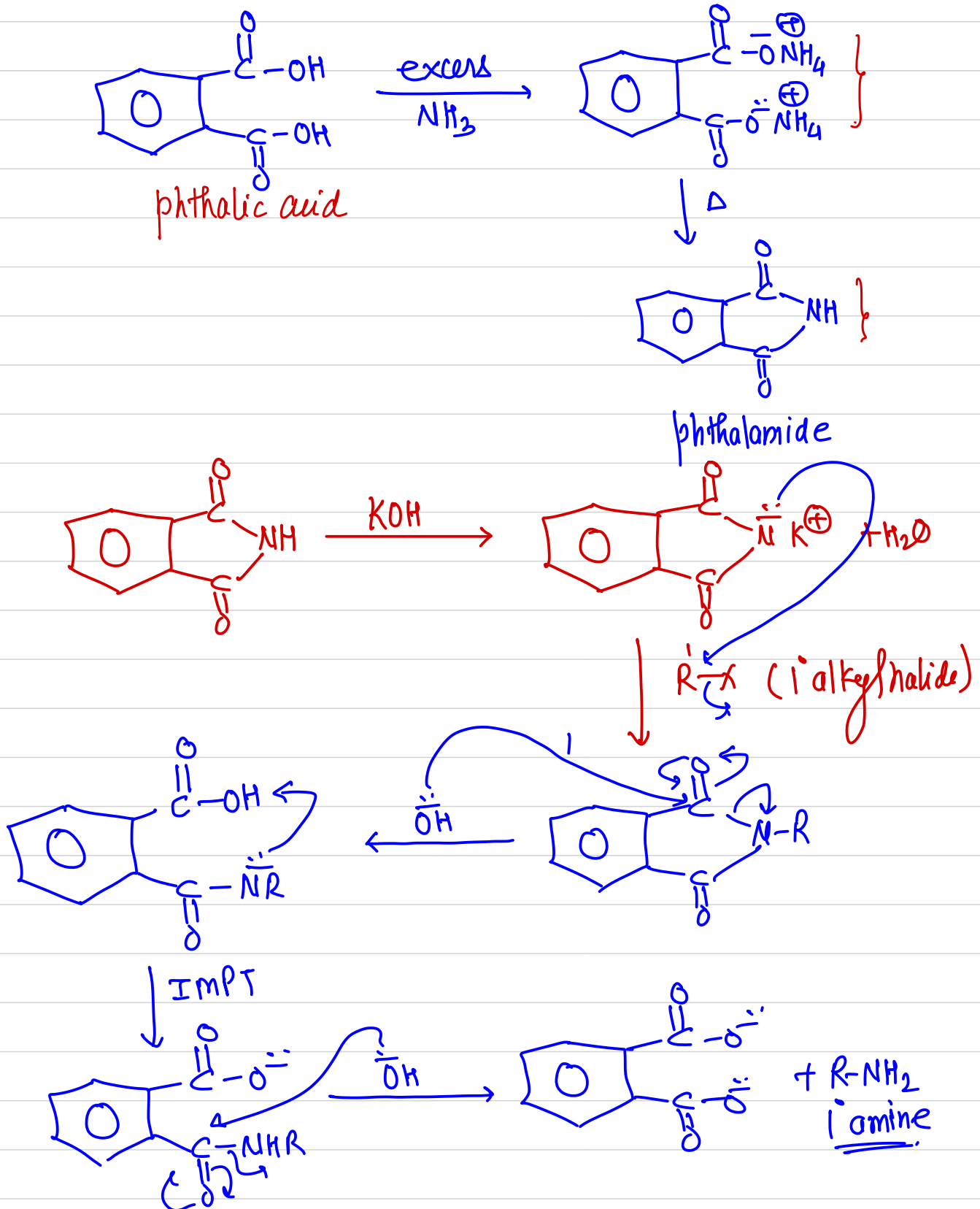


23) Wolff Kishner Reduction :-

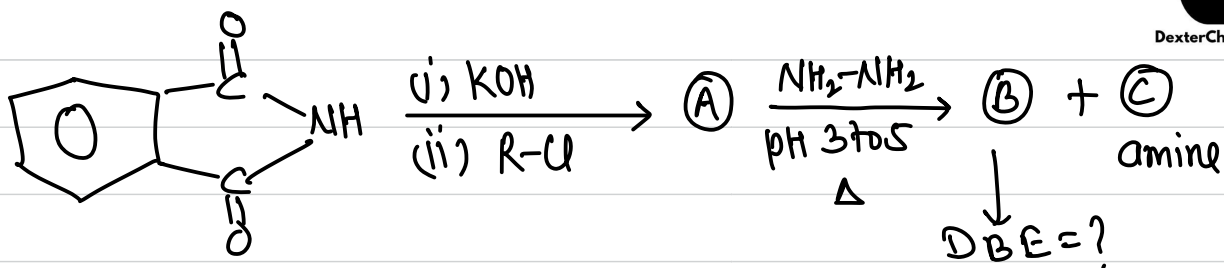


Gabriel phthalimide Reaction :-

to prepare 1° amines.

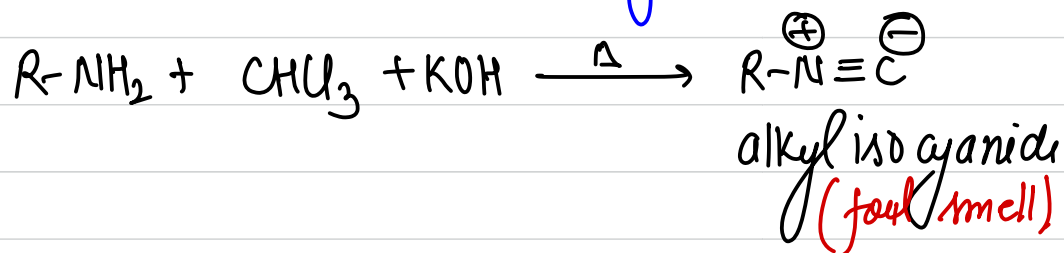


HW



Q5) Casbyl amine Reaction:-

Test of 1° amine.



Mechanism:-

